

Internal Migration and Religious Change: Evidence from Brazil's Evangelical Rise

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Abstract

This paper examines how migration affects religious markets, focusing on Brazil's sweeping religious transformation between 1980 and 2010, when Evangelicalism expanded rapidly and Catholic affiliation declined to historic lows, with far-reaching political and social consequences. Using exogenous variation in migration flows driven by international commodity price shocks, we show that migration increased Evangelical affiliation and reduced Catholic affiliation. To unpack this result, we examine two mechanisms: a demand-side channel, in which falling wages increased the appeal of Evangelicalism, and a supply-side channel, in which migration triggered a large and asymmetric expansion of Evangelical—though not Catholic—infrastructure. This asymmetry reflects deep institutional differences between Evangelical denominations and the Catholic Church. A structural model of religious choice suggests that this asymmetric supply-side response was the main driver of migration-induced Evangelical growth.

Keywords: Religion, Migration, Structural Estimation.

JEL Codes: Z12, R23, C51.

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1 Introduction

Migration reshapes societies—not just by relocating people, but by transforming the institutional, cultural, and economic environments they inhabit. While its effects on culture and politics are well documented (Collins, 2021; Alesina and Tabellini, 2024), much less is known about its impact on religious markets, despite religion’s importance in shaping identity and behavior (Barro and McCleary, 2019; Iannaccone, 1997; Iyer, 2016). This paper examines that question in the context of Brazil, where internal migration coincided with one of the most significant religious transformations of the past century: the rapid rise of Evangelicalism and the corresponding decline of Catholic affiliation. Between 1980 and 2010, the share of Evangelicals more than tripled, altering not only the country’s religious composition but also its political and social landscape (Corbi and Sanches, 2024; Costa et al., 2019; Sola, 2024; Cavalcanti et al., 2022). Similar patterns have been observed across the Global South, with far-reaching implications across multiple dimensions of social life (Pew, 2006; Hersey, 2024; Martin, 1990; Stoll, 2023).

Migration has long figured as a central explanation for Brazil’s Evangelical surge. A large body of scholarship links the growth of Evangelical churches to the massive rural-to-urban migration flows that reconfigured Brazilian cities in the second half of the 20th century; see Mariano (2011) for a survey of this literature. Crucially, at the time these flows intensified, Brazil remained overwhelmingly Catholic, especially in the rural areas from which most migrants originated.¹ Evangelicalism, by contrast, was a distinctly urban phenomenon. Its rise reflected not a diffusion of beliefs, but the emergence of new religious actors in rapidly transforming urban contexts. Yet despite its prominence in historical accounts, there is little empirical work documenting the connection between migration and Evangelical growth. This paper addresses that gap and identifies plausible channels through which migration influenced religious affiliation.

We focus not on migrants themselves, but on the communities they entered. Migration is treated as a structural shock: it expanded urban populations, strained local labor markets, and altered the institutional environment in ways that created new opportunities

¹In 1960, the Catholic share of the population was around 95%.

for religious expansion. By examining changes in affiliation among non-migrant residents, we can isolate how broader demographic pressures—rather than the religious preferences of migrants—reshaped local religious markets.² Using a shift-share instrument based on international commodity price shocks and historical migration networks, we show that migration led to a significant increase in Evangelical affiliation and a decline in Catholic affiliation among non-migrant residents.

To unpack this effect, we examine two mechanisms. First, migration depressed local incomes, which is known to increase the appeal of Evangelicalism (Costa et al., 2019; Chesnut, 1997). Second, by increasing the local population, migration expanded the potential religious market and created incentives for religious supply. Our instrumental variable estimates indicate that the number of Evangelical temples grew substantially in response to migration shocks, while the Catholic Church exhibited little or no expansion. We then develop and estimate a structural model of religious demand and find that supply-side expansion—via the growth in Evangelical temples—accounts for most of the observed religious shift, while income-driven changes in demand played a comparatively modest role.

In the first part of the paper, we combine municipality-level data on religious affiliation and migration from the 1980-2010 Brazilian Demographic Censuses with data on commodity prices and agricultural production, also measured at the municipality level. Using these data, we implement a shift-share instrumental variable design to estimate the effects of internal migration on local religious composition, focusing on decadal changes from 1980 to 2010. The instrument is constructed in two steps. First, we use international price shocks for major agricultural commodities to generate exogenous variation in out-migration from rural origin municipalities, capturing negative local income shocks that trigger displacement. Second, we allocate these predicted migrant outflows across urban destinations using pre-determined migration links from the 1980 Census (the “shares”) (Card and Lemieux, 2001). This strategy,

²Two interpretations have been particularly influential in the literature linking migration to Evangelical expansion. One centers on the migrant experience (Willems, 1967), portraying Evangelical churches as institutions that help newcomers navigate urban life by offering practical assistance, opportunities for social integration, and a clear normative framework. The other emphasizes structural change, arguing that Evangelical churches flourished not around migrants themselves but around communities reshaped by the demographic and institutional pressures generated by economic modernization—including large-scale migration (Chesnut, 1997; Novaes, 1985; Mariano, 2011). See also Hersey (2024); Ager and Ciccone (2018); Chen (2010); Murphy et al. (2022) for broader work on the integrative role of religion.

based on “push factors” as in Bazzi et al. (2023), Boustan (2010), and Derenoncourt (2022), isolates variation in migration inflows that is plausibly exogenous to religious dynamics at destination, addressing concerns about migrants’ economic sorting. Following Borusyak et al. (2022), we estimate the model in its origin-level shock-transformed form, which clarifies the source of identifying variation and ensures valid inference even when shares are correlated across destinations, as emphasized by Adao et al. (2019).³

We find that migration significantly altered religious affiliation patterns across Brazilian municipalities. Higher migration rates reduced Catholic affiliation while increasing the shares of Evangelicals and religious nones. Baseline estimates indicate that, between 1980 and 2010, migration explains a 3.7 percentage point increase in the Evangelical share, about 23% of the total observed variation in their market share.

In the second part of the paper, we explore the economic mechanisms through which migration affected Evangelical growth. We aim to assess the extent to which this shift reflects changes in religious preferences—driven by economic hardship—and/or changes in religious availability, shaped by church responses to expanding markets. A large body of research argues that Evangelicalism—particularly Pentecostalism—tends to grow in contexts of economic distress, offering material assistance, emotional support, and a sense of community where state institutions and traditional churches often fall short (Chesnut, 1997; Costa et al., 2019). At the same time, recent studies show that internal migration in Brazil has depressed wages and heightened economic insecurity, especially among informal workers (Corbi et al., 2025; Imbert and Ulyssea, 2022). Using our shift-share IV strategy, we find that migration reduced local earnings by approximately 5% on average in each decennial period covered by our data. Although not particularly strong, this result suggests that if Evangelical churches become more appealing under these conditions, then the observed religious shift may be partly explained by migration-induced wage declines.

We then turn to the supply channel. In this context, the migration shocks identified through our shift-share instrument can be interpreted as exogenous increases in the potential religious market at destination cities, generating incentives for churches to expand

³Under this approach, the estimated coefficients are numerically equivalent to those from a destination-level regression using the standard shift-share instrument (Borusyak et al., 2022).

infrastructure independently of changes in preferences. Combining Census data and detailed administrative records on the supply of Catholic and Evangelical temples across Brazilian cities, we estimate the effect of migration on religious supply. We find that, while the Catholic Church’s response was largely muted, Evangelical denominations consistently expanded their number of temples in response to migration across all decades. Back-of-the-envelope estimates suggest that, absent migration, the number of Evangelical temples per capita would have been about 27% lower—an effect far larger than the estimated impact of migration on earnings.

These findings can be rationalized by organizational differences across religious groups, as shown by Corbi and Sanches (2024). Evangelical churches operate with low fixed costs, decentralized governance, and flexible infrastructure, often relying on rented spaces and informal clergy. The Catholic Church, by contrast, depends on a centralized and capital-intensive model, requiring dedicated temples and lengthy priestly training. As a result, Corbi and Sanches (2024) show that Evangelical denominations tend to react more quickly and more intensely to positive market shocks than the Catholic Church.⁴

To assess the robustness of these results and address concerns about the validity of the instrument, we conduct a series of additional tests across all exercises. First, we include lagged values of the dependent variable to account for potential pre-trends in religious composition across municipalities. Second, we control for lagged predicted migration to mitigate concerns that our estimates may reflect persistent migration flows rather than contemporaneous shocks. Third, we also control for differential trends in economic and demographic variables—such as population size, income per capita, education, racial composition, and age structure—as well as state fixed effects, which absorb broader regional religious trends. Fourth, we account for the possibility that price shocks in origin municipalities could influence destination outcomes through other channels—such as trade or capital reallocation—by including in our models a measure of exposure of destinations to shocks in other municipal-

⁴Corbi and Sanches (2024) show that the asymmetric response of Evangelical and Catholic churches to market shocks is primarily driven by differences in entry costs—that is, the costs of opening and maintaining a temple—which, in turn, reflect deeper organizational differences between the two. These organizational features may also affect other dimensions of church behavior that help explain supply-side reactions to market shocks. For instance, Evangelical proselytism tends to be more effective than that of the Catholic Church, and Evangelical denominations are often more efficient at extracting resources from their members, which increases the expected return to opening new temples and may further amplify their responsiveness.

ities, aggregating shocks in other municipalities weighted by the inverse of the distance between destination and other municipalities. Finally, we include local food price shocks at destination as an additional control to ensure that our estimates are not confounded by local economic fluctuations that may be correlated with broader commodity cycles. While no observational strategy can eliminate all concerns about identification, the robustness of our results across multiple specifications lends support to a causal interpretation of our results.

The final part of the paper develops a structural model of religious demand in the spirit of Berry et al. (2004); see also Corbi et al. (2022) for an application to the U.S. religious market. In this model, individuals choose between the Catholic Church, Evangelical churches, or no religion, based on their income, demographic characteristics, and the religious infrastructure available in their municipality. The model captures the two key economic channels through which migration can affect religious affiliation: by altering local income levels and by changing the number of temples associated with each religious group. We estimate the model using Census microdata and find that Evangelical affiliation responds negatively to income and positively to the number of Evangelical temples. By contrast, Catholic affiliation is relatively insensitive to both income and infrastructure. More broadly, our estimates show that religious demand is considerably more responsive to changes in infrastructure than to changes in income. While we do not interpret these results as strictly causal, we show—using a range of different strategies—that they are unlikely to be driven by endogeneity in income or religious infrastructure.⁵

To gauge the relative importance of each channel, we use the estimated model to conduct two counterfactual simulations. In the first, we ask: if migration had affected only income (leaving temple supply fixed), how much would income have had to decline to generate the observed migration-driven rise in Evangelical affiliation? We find that the required income

⁵We first assess the direction of potential biases by estimating our demand model while controlling for a range of potential confounders—including religion-municipality and religion-year fixed effects—and find that the results become unequivocally stronger rather than weaker. This suggests that concerns about the endogeneity of income or infrastructure are unlikely to be driving our results. As an additional test of the infrastructure channel, we implement a staggered difference-in-differences design, following Callaway and Sant’Anna (2021), exploiting variation in the timing of temple entry across municipalities. We find that Evangelical temple entry leads to a sustained increase in Evangelical affiliation, with no evidence of pre-trends. In contrast, Catholic temple entry has no significant effect. These results reinforce the interpretation that Evangelical infrastructure expansion played a key role in driving religious change.

reduction would need to be between 60% and 80%—an implausibly large drop given the actual effects of migration on income estimated in the data. In the second simulation, we hold income fixed and ask: if migration had affected only the number of Evangelical temples, by how much would supply have to expand to reproduce the observed religious shift that was induced by migration? The answer—about a 20% increase in temples—closely matches the actual migration-induced expansion in Evangelical infrastructure. Ultimately, these results indicate that Evangelical growth appears to be driven more by differences in institutional adaptability to migration than by differences in how income affects affiliation.⁶

This paper makes four main contributions. First, it provides empirical evidence that internal migration—and more broadly, population growth—can be a major driver of religious change. While anecdotal and descriptive accounts often invoke migration as a background factor, we establish a link between migration shocks and changes in local religious composition. We show that migration led to increased Evangelical affiliation and a decline in Catholic affiliation among non-migrant residents, with temple construction being an important determinant of this effect. We believe that this result is relevant to a vast literature studying the origins of Evangelicalism in many parts of the world, particularly Latin America (Martin, 1990; Stoll, 2023).

Second, we also contribute to the long-standing debate over what explains the rise of Evangelicalism in Brazil and elsewhere. The dominant view links this growth to poverty and exclusion, arguing that Evangelical churches attract the poor by offering emotional support, behavioral discipline, and community where state and traditional institutions fail (Mariano, 2011; Chesnut, 1997; Martin, 1990; Costa et al., 2019; Cavalcanti et al., 2022). But a growing literature emphasizes supply-side factors, particularly organizational differences, as key to understanding religious expansion (Finke and Stark, 1988; Stark and Finke, 2000; Iannaccone et al., 1997; Stark and Iannaccone, 1994); see also Putnam et al. (2012). In Brazil, Evangelical churches have adopted a decentralized, low-cost model based on informal leadership and rented storefronts (Corbi and Sanches, 2024), while the Catholic Church

⁶As we are not explicitly modelling the supply of temples, this exercise abstracts from potential indirect effects of income on Evangelical affiliation operating through changes in temple supply. In any case, we view this as a minor limitation. Given the weak sensitivity of demand to income, such effects are likely to be small and do not affect our main conclusions.

remains capital-intensive and slow-moving, requiring purpose-built temples and years of clergy training (Mariano, 2004). These institutional asymmetries shape each group’s capacity to respond to population shocks. By showing that migration-driven market expansion led to asymmetric infrastructure growth and religious conversion, our paper provides empirical support for the supply-side explanation.

Third, we contribute to a still incipient literature on how migration fosters transformations in religious markets. Bazzi et al. (2023) show that the migration of white Southerners helped spread Evangelical churches and conservative political values to other regions of the United States. Hersey (2024), by contrast, emphasizes the role of Evangelicals in supporting migrants in Ghana. While the former focuses on the cultural transmission of religious and political preferences by migrants, the latter highlights the experience of migrants themselves. Our approach centers on a different mechanism: how internal migration transforms religious markets not through belief diffusion or migrant experience, but by altering local economic conditions, prompting supply-side responses that reshape the behavior of native populations.

More broadly, we contribute to a growing body of empirical work in the economics of religion that examines how structural factors contribute to religious change. Barro and McCleary (2005) show that state sponsorship of religion increases participation, while greater government regulation reduces both religious attendance and belief. Gruber and Hungerman (2008) find that repealing Sunday trading restrictions (“Blue Laws”) led to a decline in church attendance, while Hungerman (2005) documents substitution between government programs and church-based activities. Cantoni et al. (2018) show how the Reformation reconfigured religious markets across German cities, and Bazzi et al. (2020) demonstrate that the transfer of resources from rural elites to Islamic institutions was instrumental in consolidating Islam’s political influence in Indonesia. Unlike previous work emphasizing state policy or historical legacies, we show that internal migration in Brazil reshaped the religious landscape through institutional asymmetries in how churches respond to demographic change.

The paper is organized as follows. Section 2 describes the institutional context and the data used throughout the analysis. Section 3 outlines the empirical strategy and presents estimates linking internal migration to the rise of Evangelicalism. Section 4 investigates the mechanisms behind this relationship. Section 5 develops a structural model of religious de-

mand and uses it to quantify the relative importance of each mechanism. Section 6 concludes the paper.

2 Institutional Background and Data

This section presents the primary datasets used in our analysis and offers an overview of Brazil’s religious landscape. It focuses on the recent transformation of the country’s religious markets, marked by the rapid expansion of Evangelicalism and the concurrent decline of Catholic affiliation. We also highlight some of the visible social consequences of this shift, illustrating that its effects extend well beyond the boundaries of religious markets.

2.1 Data Sources

The data utilized in this paper is derived from several sources. The first is the Brazilian Census, which provides information on religious affiliations and migration patterns. Our study incorporates data from the 1980, 1991, 2000, and 2010 Census waves, the 2010 Census being the most recent available.

The cross-sectional unit of analysis in our study is the municipality, defined according to the administrative boundaries in place in 1980, the first year of our sample. Since then, numerous municipalities have undergone border changes, often due to subdivisions. To maintain comparability over time, we reassign all newly created municipalities to the original municipalities from which they were formed.⁷

In the Brazilian Censuses of 1980, 1991, 2000, and 2010, respondents were asked, “What is your religion?” with a predefined list of options that evolved to reflect the country’s increasing religious diversity. Across all waves, these options allow us to categorize respondents into broad religious groups, including Catholic, Evangelical Mainline, Evangelical Protestant (encompassing Pentecostal, Non-Pentecostal, and Non-denominational Evangelicals), Spiritist, Afro-Brazilian religions (e.g., Umbanda, Candomblé), Jewish, “Other”, and “No Religion.”

⁷Using shapefiles for municipal boundaries in 1980 and 2010, we identified the centroid coordinates of each 2010 municipality and overlaid them onto the 1980 polygons to determine their corresponding original municipalities.

Throughout this paper, we group Evangelical Mainline and Evangelical Protestant denominations under a single category: “Evangelicals” (see Table A1 in the Appendix). Our primary classification thus consists of two main groups: Catholic and Evangelical.⁸ Other religious groups—such as Spiritist, Afro-Brazilian, and non-Christian religions—remain consistently small, accounting for less than 5% of the population. Based on this classification, we construct measures of religious affiliation that serve as key dependent variables in our analysis.

Additionally, in the Brazilian Censuses of 1980, 1991, 2000, and 2010, respondents were asked about their place of residence before the Census date. We use this information to construct (i) migration inflows for each municipality, which serve as the primary independent variable in our analysis, and (ii) “past settlements” at the municipality level, which form the “share” of the shift-share instrumental variable (SSIV) approach used to address the endogeneity of migration inflows; see Section 3 for details.

Specifically, in the 1980, 1991, and 2010 Brazilian Censuses, migration-related questions included “How long have you lived in this municipality?” and “What municipality did you live in before?” However, the latter question only applied to individuals who had lived in their current municipality for ten years or less and is missing otherwise. In the 2000 Census, the questions were “How long have you lived in this municipality?” and “What municipality did you live in five years ago?”—with answers to the latter question missing for individuals who had moved to their current municipality more than five years prior. Given these differences, to ensure a homogeneous measure of migration rate across Census waves, we define a migrant as an individual who moved to their current municipality of residence (destination) within the past five years; see also Imbert and Ulyssea (2022), Corbi et al. (2025). The origin municipality is defined as the municipality where the individual previously lived, based on the Census responses.

In addition to data on religious affiliation and internal migration flows, we collected economic and demographic variables from the Brazilian Census, which serve as controls in some of our regressions. Among these, household average income is of particular importance,

⁸For details on the specific response options across Census waves, see the IBGE documentation on religion in the Brazilian Census, available at <https://www.ibge.gov.br/>.

as it functions as the dependent variable in parts of the analysis where we examine the mechanisms underlying our results.

To further investigate mechanisms, we collected data on the supply of religious temples from the Brazilian Internal Revenue Service (BRASIL, 1991-2018), which maintains a comprehensive registry of all legal entities in the country. We retain all entries classified as Religious Organizations under the Brazilian Industry Classification (CNAE). The dataset provides establishment-level information, including start and end dates, address, and entity name. Based on this information, we classify each religious establishment into one of three groups: Catholic, Evangelical, or Other Religions. We then reorganize the data into a municipality-year panel, in which each observation reflects the active stock of temples for each religious group in a given year and municipality. This variable serves as the main outcome in our analysis of supply-side responses to migration shocks.

Finally, we collected data on commodity prices and agricultural production at the municipality level in Brazil. This information is used to construct the “shifters” in our shift-share instrumental variable (SSIV) strategy, as detailed in the next section. Following Imbert and Ulyssea (2022), we use monthly price data for 12 commodities—banana, cocoa, coffee, cotton, maize, orange, rice, soybeans, sugar, tobacco, wheat, and wood—sourced from the World Bank’s Pink Sheet, covering the period from 1970 to 2010. Using these data, we construct commodity-specific price shocks, which are then aggregated at the origin municipality level using the value share of each crop in the total agricultural output of that municipality.

To account for the relative importance of agriculture in each municipality, we also weight these shocks by the share of local agricultural production in the national total. This information is obtained from the Agricultural Census and the Municipal Agricultural Survey, both published by IBGE. Further methodological details on how we combine commodity prices and local production shares to build our SSIVs are provided in the next section.

This concludes the description of our data sources and variables. In the remainder of this section, we use parts of this data to provide a panorama of the Brazilian religious context and highlight patterns that are particularly relevant to our analysis.

2.2 The Brazilian Religious Market

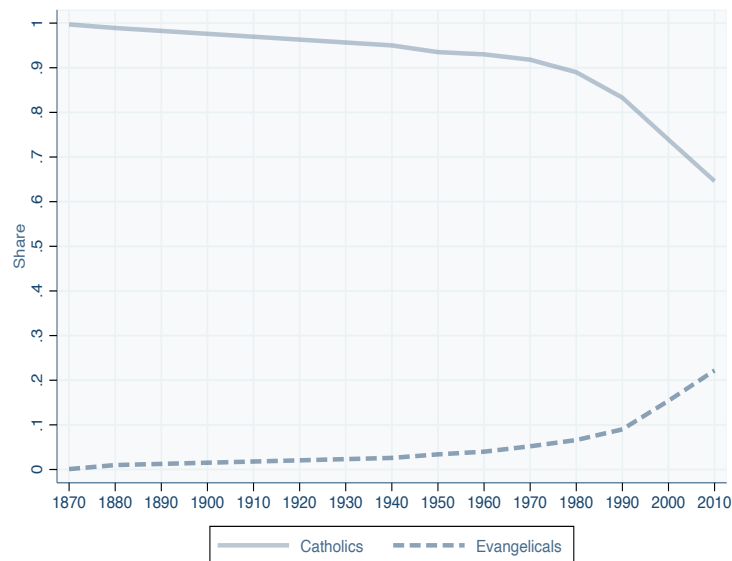
Brazil's religious landscape has been historically shaped by its colonial past and subsequent transformations, which began in the early 20th century and became critical during the final decades of that century. During Portuguese colonization, Catholicism was established as the official religion, deeply intertwined with the state's governance and cultural life. Missionary efforts by Jesuits and other Catholic orders played a central role in converting Indigenous populations and integrating enslaved Africans into the colonial religious framework. By the time of Brazil's independence in 1822, Catholicism was so dominant that over 99% of the population identified with the faith. Even after the formal separation of church and state in 1891, Catholicism retained its cultural and social prominence, influencing traditions, moral norms, and public life well into the 20th century (Scott, 1989).

Despite Catholic dominance, Brazil has always had a diverse religious landscape, which became increasingly pluralistic with the expansion of Protestantism in the 19th and 20th centuries. Mainline Protestantism, introduced by European and North American missionaries, established itself through denominations such as Lutheran, Anglican, and Calvinist churches. Non-Pentecostal Evangelical groups also emerged around the same time, largely founded by U.S. and European missionaries, and include the Baptist, Methodist, and Adventist churches (Chesnut, 1997; Mariano, 2014). These traditions emphasized structured worship, doctrinal teaching, and social engagement, often attracting more urban and middle-class followers (Freston, 2001).

Pentecostalism emerged in the early 20th century with the establishment of the Christian Congregation in 1910 and the Assemblies of God in 1911 (Chesnut, 1997). Rooted in the global Pentecostal revival, Pentecostalism emphasizes personal experiences with the Holy Spirit, such as speaking in tongues and divine healing. Known for their emotionally charged worship and rapid grassroots expansion, Pentecostal churches effectively penetrated poor urban neighborhoods and rural areas, offering, according to the existing literature, a sense of belonging and hope to marginalized populations (Chesnut, 1997; Mariano, 2014). Despite their differences, the term "Evangelicals" is often used to refer to the combined group of Mainline Protestants, Non-Pentecostal and Pentecostals.⁹

⁹Some authors (Mariano, 2014) further distinguish Pentecostals into two categories: traditional Pente-

Figure 1: **Shares of Evangelicals and Catholics in the Brazilian population (1870-2010)**



This figure shows the shares of Catholics and Evangelicals (including Mainline, Pentecostals and Non-Pentecostals). These are the main religious groups in Brazil. Data obtained from Alves et al. (2014), originally sourced from the Brazilian Censuses of 1870, 1880, 1940, 1950-1980, 1991, 2000, 2010, conducted by IBGE (Instituto Brasileiro de Geografia e Estatística). Data for 1890-1930 interpolated using linear interpolation.

Other traditions, such as Afro-Brazilian religions brought to Brazil by enslaved Africans, Spiritism (notably influenced by Allan Kardec’s teachings), and non-Western religions, maintain small but persistent representation in the country’s religious mosaic.

More recently, Brazil’s historically established religious landscape has undergone profound changes. The most notable of these shifts is the decline in Catholic affiliation and the rapid rise of Evangelicals. Figure 1 illustrates the magnitude of these movements. According to data from the Brazilian Censuses, the proportion of the population identifying as Catholic fell from approximately 90% in 1980 to 65% in 2010—a 25 percentage point decline over 30 years. To emphasize the pace of this change, consider that over the 100 years between 1870 and 1970, the share of Catholics had decreased by only about eight percentage points. During the same 1980–2010 period, the share of Evangelicals increased dramatically

costals, which include denominations that arrived early (such as the Assemblies of God, Christian Congregation, and later the Foursquare Church) or were founded in the 1950–60s (e.g., Brazil for Christ, God is Love), and neo-Pentecostals, which encompass churches established in the late 1970s, such as the Universal Church of the Kingdom of God and the International Grace of God Church.

from 6% to 22%, a remarkable three- to fourfold increase, with the sharpest acceleration occurring between 1991 and 2000, a decade during which the share of Evangelicals grew by an astonishing 71%.

Due to its rapid pace and significance, this phenomenon has had profound implications across different spheres of Brazilian society, most notably in politics (Freston, 2001; Costa et al., 2019; Corbi and Sanches, 2024; Sola, 2024; Smith, 2019; Araújo, 2022). Historically, apart from isolated cases, the participation of Evangelical groups in politics began gaining prominence during the Constituent Assembly of 1987, convened to draft a new Brazilian Constitution following the end of the military dictatorship (1964–1985).

Since their initial forays into political power, Evangelical positions have consistently emphasized the defense of Christian values, particularly through opposition to abortion and LGBTQ+ rights (Costa et al., 2019; Freston, 2001), alongside the enforcement and expansion of the economic agenda of religious institutions, which primarily includes tax exemptions for religious organizations and concessions of radio and TV networks—typically used for proselytism (Freston, 2001). More recently, the alignment of Evangelical churches with far-right parties has become explicit, with many prominent Evangelical leaders openly endorsing Brazil’s former right-wing president Jair Bolsonaro and his supporters during the last four elections (2018, 2020, 2022, and 2024)—being decisive in Bolsonaro’s first election and in the elections of State Governors and Mayors of the most important Brazilian cities.

At a deeper level, some authors (Araújo, 2022) have argued that, through their influence over the masses, Evangelical leaders have reshaped the traditional voting behavior of lower-income groups in Brazil. Historically, this demographic’s vote was primarily driven by economic performance and redistributive policies. However, there is growing evidence that Evangelical voters increasingly support candidates whose moral agendas align with their religious values; see also Retzl (2025) and Smith (2019). This shift has moved the political sympathies of the poorer and lower middle classes away from left-wing parties, which typically advocate for redistributive policies, toward right-wing parties, which are more aligned with conservative moral values. Overall, the religious transition has had profound consequences for Brazilian society.

3 Internal Migration and the Rise of Evangelicals

The previous section illustrated the pace and depth of Brazil’s religious transition and its political implications. This section formally examines the role of migration flows in shaping these changes, particularly their contribution to the rise of Evangelicalism. Internal migration, which accelerated in the 1960s and continued through the 1990s, played a pivotal role in reshaping urban communities by relocating millions from rural areas to rapidly expanding cities. By the end of this period, approximately 35% of the Brazilian population resided in a city different from their birthplace, according to IBGE data. These large-scale migrations not only transformed the urban demographic landscape but also created fertile ground for religious shifts, particularly the expansion of Pentecostal and Evangelical movements.¹⁰

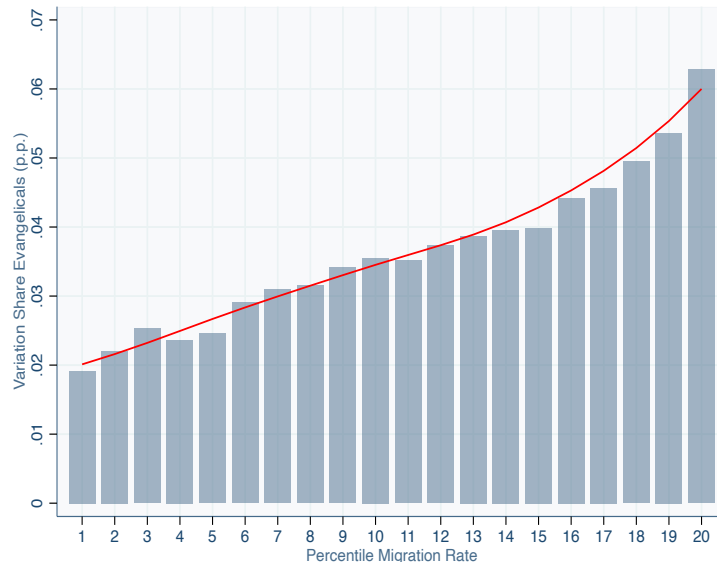
While much of the existing literature has focused on the religious conversion of migrants (Mariano, 2011; Hersey, 2024)—emphasizing how newcomers to urban centers often seek support within Evangelical churches, particularly in areas where state assistance is limited—*this study shifts the focus to how migration reshapes the religious conversion of native residents*. This perspective is crucial because migration is not an isolated event affecting only those who move; rather, it is a disruptive force that transforms entire local economies, neighborhoods, and institutions, ultimately influencing the religious choices of those who never relocated. Internal migration has been shown to affect Brazilian labor markets, leading to wage declines in both the short and long term (Corbi et al., 2025; Imbert and Ulyssea, 2022). These economic downturns, in turn, have been linked to a rise in conversions from Catholicism to Evangelicalism (Costa et al., 2019).

Another channel through which migration may lead to the conversion of non-migrants in urban centers is the expansion of religious infrastructure. The influx of migrants often necessitates the establishment of new religious institutions to serve these communities. This increase in religious infrastructure enhances the visibility and accessibility of alternative religious practices, thereby influencing the religious affiliations of native residents. Evangelical churches, in particular, are known to be more flexible and responsive to demand shocks of

¹⁰For anecdotal discussion on the relationship between migration and the Pentecostal upsurge, see <https://veja.abril.com.br/coluna/maquiavel/tres-fatores-que-explicam-o-fenomeno-do-boom-evangelico-no-brasil>.

this nature than the Catholic Church (Corbi and Sanches, 2024). This suggests that the opportunities created by market size expansions may have been asymmetrically captured by Evangelical churches. In Section 4, we formally investigate the role of these channels in shaping the effects of migration on the religious choices of the native population.

Figure 2: Migration Rates Across Cities and Variation in the Share of Evangelicals among Native Residents



The graph illustrates the relationship between in-migration rates and the variation in the share of Evangelicals among native residents across Brazilian municipalities. Here, native residents are defined as the total population of a municipality, excluding individuals identified as migrants, following the IBGE's conceptualization discussed in the previous section. The x-axis represents percentiles of migration rates, ranging from the 1st percentile (municipalities with the lowest migration rates) to the 20th percentile (municipalities with the highest migration rates). The bars depict the average share of Evangelicals within each percentile, while the line represents the LOWESS (Locally Weighted Scatterplot Smoothing) smoothed series. Data source: Brazilian Census from 1980–2010.

Figure 2 provides descriptive evidence supporting the hypothesis that internal migration altered the composition of Brazil's religious landscape, particularly fostering Evangelical growth among native (non-migrant) residents of Brazilian cities. The horizontal axis represents percentiles of (in)migration rates—calculated as the number of (in)migrants in a municipality divided by the total local population—while the vertical axis shows the average variation in the share of Evangelicals in native populations for each percentile. The strong upward trend in the figure indicates that municipalities with higher migration rates experienced significantly greater Evangelical growth among native residents.

The descriptive evidence is consistent with the idea that internal migration contributed

to the growth of Evangelicalism in urban areas. To move beyond correlations, the remainder of this section develops a causal interpretation of the relationship between migration and the rise in Evangelical affiliation, as shown in Figure 2. We use a shift-share instrumental variable (SSIV) strategy that exploits exogenous variation in commodity price shocks, weighted by historical agricultural production in each municipality, to instrument for migration rates. The following subsections describe this identification strategy in detail and explain how it is implemented. Once the causal effect of migration on Evangelical growth is established, we turn to the mechanisms driving this relationship.

3.1 Empirical Strategy

To estimate the effects of migration on the share of *native* populations affiliated with different religious groups in a destination municipality d , we specify the following regression model (for simplicity, we omit time subscripts from the specification):¹¹

$$\Delta s_d = \alpha + \beta m_d + \gamma X_d + \epsilon_d, \quad (1)$$

In this specification, s_d denotes the share of the native population affiliated with a given religious group (e.g., Catholic or Evangelical) in a given Census year. The variable m_d captures the accumulated inflow of migrants into destination d during the period covered by that Census wave. The vector X_d includes baseline controls at the destination level, and ϵ_d is the error term. We estimate separate regressions for the 1991, 2000, and 2010 Census waves.

The coefficient β in equation (1) captures the overall effect of migration on Δs_d , operating through its influence on factors shaping both the demand and supply of religion. Therefore, it does not represent the direct causal effect of migration on Δs_d but rather a reduced-form parameter that summarizes migration’s combined impact through different economic mechanisms.

¹¹In practice, since we estimate the models separately for each period—1980-1991, 1991-2000, and 2000-2010—the time subscript becomes redundant. We adopt this approach because, as we will see later, the relationship between variables appears to change across periods, and we aim to ensure that our model is flexible enough to capture these shifts in the data.

In this section, we remain agnostic about the specific mechanisms driving the estimated effect and focus on identifying and testing the reduced-form parameter β . The next section examines how different channels contribute to the composition of β , providing a deeper understanding of its underlying components. Finally, the last section introduces a structural model that explicitly quantifies the role of each channel in shaping the overall impact of migration on the religious composition of the population.

Ordinary Least Squares regressions based on specification (1) may not identify β because migration is itself an endogenous decision. Migrants may select specific destination municipalities in response to local economic conditions, such as positive demand shocks that increase wages or improve job prospects. According to the literature, areas experiencing such economic gains tend to exhibit lower rates of conversion to Evangelicalism (Costa et al., 2019; Chesnut, 1997). This selective migration pattern raises concerns about potential endogeneity in OLS estimates. To mitigate this issue, we employ a first-differences approach, which removes time-invariant unobserved characteristics that might be correlated with migrant inflows. However, this method does not account for time-varying confounders, which could still bias the results.

To address these considerations, we adopt a shift-share instrumental variable (SSIV) approach. In the first step, we use commodity price shocks at origin municipalities to predict migration outflows (the “shift”). The underlying idea is that municipalities experiencing negative price shocks to the commodities they produce are more likely to generate a larger outflow of migrants compared to those facing positive price shocks. In the second step, we allocate the predicted aggregate migrant supply to destination municipalities based on pre-existing settlement patterns (the “share”); i.e., we assume that migrants are more likely to relocate to destinations where a larger population from their origin already resides, reflecting the role of social networks and established community ties in shaping migration flows as in Card and Lemieux (2001).

We construct a measure of agricultural income shocks for origin municipalities, drawing on the approach of Imbert and Ulyssea (2022). Specifically, for each crop c and month l in year y , we define a price shock, ε_{cly} , as the residual from an AR(1) process applied to international commodity prices. These shocks are then aggregated at the municipality level—denoted by

o —using two weights: (i) the share of crop c in the total agricultural production value of municipality o , π_{oc} ; and (ii) the proportion of municipality o 's agricultural harvested area relative to its total area as of 1980, h_o . The latter weight assigns greater significance to price shocks in rural areas where agriculture plays a pivotal role in the local economy. Formally, price shocks at municipality o and year y , z_{oy}^p , is computed as:

$$z_{oy}^p = \sum_l \sum_c (\pi_{oc} \times h_o \times \varepsilon_{cly}). \quad (2)$$

We then use the aggregated price shocks to predict migration outflows according to the following process:

$$m_{oy} = \kappa + \eta z_{oy-1}^p + \phi_o + \delta_y + \xi_{oy} \quad (3)$$

where, m_{oy} is the out-migration rate from origin $o \in O$ — O is the set of origins—at year y ; z_{oy-1}^p is the price shock that hits this origin municipality in the previous year, as defined by equation (2); ϕ_o and δ_y are municipality and year fixed effects, respectively; and ξ_{oy} is a random error term. We pool the 1991, 2000, and 2010 Census waves and construct a panel of yearly out-migration from each possible origin to estimate this equation. For each year, the predicted number of migrants who leave their hometowns is obtained by multiplying this predicted rate by the municipality population reported in the 1980 Census:

$$\widehat{M}_{oy} = \widehat{m}_{oy} \times P_o. \quad (4)$$

In the second step, we use the past settlements of migrants from the origin o to municipality d to distribute price shocks throughout the destination areas, defining our SSIV as

$$\widehat{M}_{dy} = \sum_{y-5 < t \leq y} \left(\sum_{o=1}^O \nu_{od} \times \widehat{M}_{ot} \right) \quad (5)$$

where $y = \{1991, 2000, 2010\}$ and ν_{od} represents the share of migrants from origin municipality o who resided in destination area d in 1980.¹² To facilitate the interpretation of the

¹²As discussed in the previous section, we standardize the concept of migration across Census waves by

estimated coefficients, we divide the predicted number of incoming migrants by the native population in each destination municipality in 1980, P_d ; this is a simple normalization and it does not play any role in the identification of β . Therefore, the SSIV for the observed incoming migration rate in destination municipality d is:

$$\hat{m}_{dy} = \frac{\hat{M}_{dy}}{P_d}. \quad (6)$$

The credibility of the shift-share instrumental variable approach hinges on assumptions related to the shocks, the exposure shares, or both. Recent literature has formalized these conditions and examined their implications. Goldsmith-Pinkham et al. (2020) highlight that consistency can be ensured if the exposure shares are strictly exogenous. An alternative perspective, developed by Borusyak et al. (2022), suggests that identification can instead rely on the exogenous component of the shocks. Building on the framework of Borusyak et al. (2022), we exploit the exogeneity of economic shocks at the origin to identify the causal impact of migration rates on the outcomes of interest, captured by the parameter β in equation (1). Under this perspective, only the exogeneity of commodity prices at origins is critical for identification.

Our identification strategy assumes that commodity price fluctuations in origin regions are uncorrelated with unobserved factors influencing religious choices in destination areas. A key concern is that existing literature indicates that economic conditions significantly impact religious conversion (Costa et al., 2019). One plausible channel through which shocks at the origin may affect the destination’s economy is through trade; for example, destination economies may process agricultural production from the origin areas. Similarly, an increase in commodity prices could affect food prices in the destination, potentially influencing our outcome variables. We show that our results are robust to these concerns. We also demonstrate that our findings hold when accounting for pre-trends in the outcome variables, temporal correlations in migration, and by including controls for varying trends in economic and demographic characteristics across destinations. Some alternative specifications also incorporate state dummies to address different growth trajectories of the dependent variable

considering individuals who arrived at the destination at most five years before the Census interview. This explains the summation over t between y and $y - 5$.

across states.

For inference on β , we follow Borusyak et al. (2022) and transform the estimation into an origin-level regression. This yields the same point estimates as the destination-level specification—consistent with their identification result—and we use the standard deviation of the coefficient from the origin-level regression to construct standard errors and conduct hypothesis testing.

This approach also addresses concerns raised by Adao et al. (2019), who show that standard inference in shift-share regressions may be invalid when exposure shares are correlated across units, leading to downward-biased standard errors and overrejection of the null hypothesis. As Borusyak et al. (2022) demonstrate, conducting inference at the level of shocks (i.e., using origin-level variation) produces standard errors that are asymptotically equivalent to those obtained via the correction procedure proposed by Adao et al. (2019).

3.2 Results

Before presenting the main results, we first examine the outcomes of our first-stage regressions, represented by equation (2). Table 1 displays the results of these regressions under various model specifications. The first two columns present the coefficients from the regression of migration outflows from each municipality on our measure of commodity price shocks. In the third column, we extend the analysis by interacting price shocks with dummies for positive and negative shocks. All specifications include time and municipality fixed effects. Columns 2-3 also include baseline covariates (share of population under 18 years old; share of non-white; share of female; share with high school education or higher; share of married individuals; household income per capita, and total population) measured in 1980 and interacted with time dummies. Standard errors in parentheses are clustered at the municipality level.

In specifications 1 and 2, the coefficients are negative and statistically significant at the 1% level, indicating that negative shocks to the prices of agricultural products produced in municipality o lead to an increase in (out)migration from that municipality to other areas. The inclusion of covariates has little effect on the point estimates. The third column further reveals that the effect of price shocks on migration is highly asymmetric, with migration

responding much more strongly to negative shocks than to positive ones.¹³

One potential concern is that price shocks for certain commodities—particularly soy—may induce structural changes in local economies (Bustos et al., 2016), thereby confounding their relationship with out-migration flows.¹⁴ To address this, we re-estimate the specifications, excluding soy from the commodity price basket. As shown in columns 4–6, the results remain very similar.

Table 1: **Migration Outflows Induced by Price Shocks**

	(1)	(2)	(3)	(4)	(5)	(6)
Price shocks _{<i>y</i>-1}	-0.005*** (0.001)	-0.004*** (0.001)				
Negative shocks _{<i>y</i>-1}			-0.007** (0.004)			
Positive shocks _{<i>y</i>-1}			-0.000 (0.003)			
Price shocks _{<i>y</i>-1} (No soy)				-0.004*** (0.001)	-0.002** (0.001)	
Negative shocks _{<i>y</i>-1} (No soy)						-0.005* (0.003)
Positive shocks _{<i>y</i>-1} (No soy)						0.001 (0.003)
F Statistic	11.8	28.2	28.5	9.2	28.6	29.3
Observations	58,905	58,905	58,905	58,905	58,905	58,905
Municipalities	3,927	3,927	3,927	3,927	3,927	3,927
Time dummies	✓	✓	✓	✓	✓	✓
Municipality dummies	✓	✓	✓	✓	✓	✓
Baseline covariates		✓	✓		✓	✓

Each observation is an origin municipality by year. The dependent variable is the number of individuals who left the origin area, divided by its total population in 1980. All specifications include time and municipality fixed effects. Columns 2-3 and 5-6 also include baseline covariates (share of population under 18 years old; share of non-white; share of female; share with high school education or higher; share of married individuals; household income per capita, and total population) measured in 1980 and interacted with time dummies. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

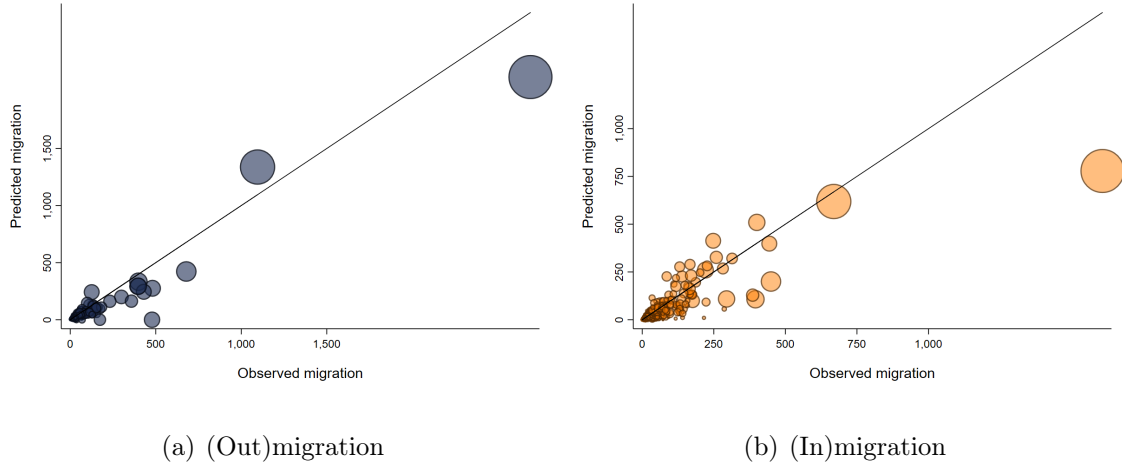
To provide further support for our first-stage regressions, Figure 3 plots observed migration outflows in each municipality alongside those predicted by the model in column 1

¹³In principle, both negative and positive price shocks at origin municipalities may induce migration outflows. While negative shocks can force individuals to seek better economic conditions elsewhere, positive shocks can ease financial constraints for those who were already considering migration, allowing them to pursue opportunities in other regions (Bazzi, 2017).

¹⁴Soy is by far the most economically significant commodity in our baseline specification.

of Table 2 (Panel (a)), as well as observed migration inflows and model predictions (Panel (b)). Observations are weighted by municipality population in 1980, with the size of each circle proportional to these weights. Overall, the figure indicates that our migration model provides a strong fit for both migration inflows and outflows, suggesting a robust first-stage for our IV estimates.

Figure 3: **Predicted vs. Observed Migration Flows**



This figure illustrates the relationship between predicted and observed migration flows across Brazilian municipalities from 1985 to 2010. Panel (a) depicts the number of migrants leaving their hometowns, while Panel (b) presents the number of incoming migrants in destination areas. The size of each circle corresponds to the total population of the municipality in 1980. Data source: Brazilian Census.

We now turn to the second-stage results, which examine the relationship between migration rates and changes in the religious composition of native residents within each municipality. Table 2 summarizes these results. Columns report estimates from both OLS and SSIV regressions, with outcomes corresponding to changes in the share of Catholics (columns 1 and 2) and Evangelicals (columns 3 and 4). Panel A covers the period 1980–1991, Panel B covers 1991–2000, and Panel C covers 2000–2010. All regressions include the baseline covariates described earlier. SSIV coefficients and standard errors (in parentheses) are estimated using origin-level regressions, with standard errors clustered at the municipality-of-origin level (Borusyak et al., 2022).

The first-stage F-statistics, shown at the bottom of Panels A–C, are very large, varying between 199.1 and 68.4.¹⁵ These values confirm the strong first-stage effects discussed ear-

¹⁵As shown by Borusyak et al. (2022), in a shock-level SSIV setup, the F-statistic is calculated as the square of the t-statistic of the first-stage coefficient for the same sample used in the second-stage.

Table 2: **Effects of Migration on the Share of each Religious Group among Native Populations**

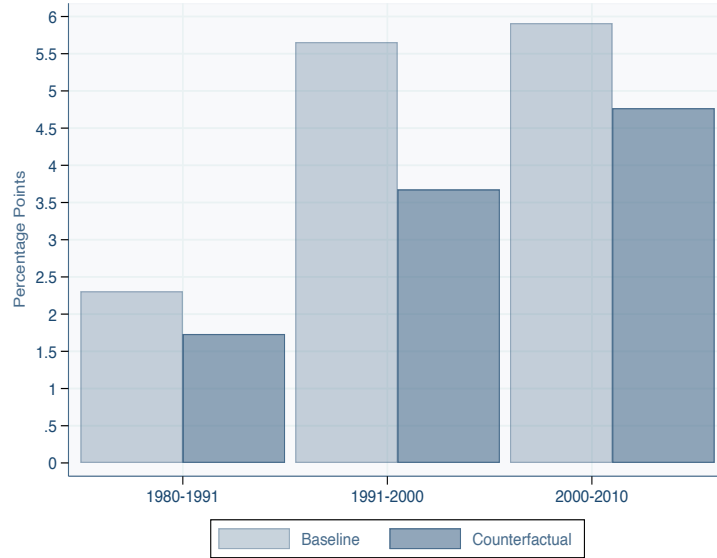
	Dep. Var.: Δ share of religious group			
	Catholics		Evangelicals	
	(1)	(2)	(3)	(4)
	OLS	SSIV	OLS	SSIV
<u>Panel A: Δ 1980-1991</u>				
In-migration rate	-0.023* (0.014)	-0.109* (0.063)	0.018** (0.007)	0.046* (0.025)
First Stage F Stat		186.4		181.6
Observations	3,919	3,989	3,919	3,989
<u>Panel B: Δ 1991-2000</u>				
In-migration rate	-0.050*** (0.011)	-0.167*** (0.020)	0.039*** (0.007)	0.148*** (0.030)
First Stage F Stat		199.1		186.6
Observations	3,919	3,989	3,919	3,989
<u>Panel C: Δ 2000-2010</u>				
In-migration rate	-0.032*** (0.005)	-0.103*** (0.014)	0.026*** (0.004)	0.073*** (0.010)
First Stage F Stat		69.0		68.4
Observations	3,919	3,989	3,919	3,989
Baseline covariates	✓	✓	✓	✓
Lagged religion share	✓	✓	✓	✓

This table shows municipality of destination-level OLS and municipality of origin-level SSIV coefficients on the 5-year accumulated changes in the municipality-level share of each religious group among native individuals. The SSIV is the outflow of migrants predicted by price shocks at the origin municipality. In Panel A, the dependent variation is the change in outcome between 1980 and 1991; in Panel B, between 1991 and 2000, while in Panel C, it is the change in the outcome between 2000 and 2010. All regressions include controls for lagged covariates (share of population less than 18 years old; share of non-white; share of female; share with complete high-school or higher; share of married people in the population; household income per capita, and the total population) as well as the share of the respective religious group in the base year. Estimates are weighted by the total population in 1980. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

lier in this section. Comparisons between OLS and SSIV estimates suggest that the OLS estimates are systematically biased toward zero. One plausible explanation for this bias is that migrants tend to select regions with stronger economic performance, where the Catholic Church performs particularly well, while Evangelical churches perform relatively worse. This mechanism, discussed in Costa et al. (2019), aligns with the observed patterns and highlights the importance of addressing endogeneity in this analysis.

Our analysis of the SSIV estimates in Table 2 reveals two key findings. First, higher migration rates consistently lead to a decline in the share of Catholics across all periods analyzed. Second, this reduction in Catholic affiliation is accompanied by a statistically significant increase in the share of Evangelicals, with the effect being particularly pronounced during the 1991–2000 period.

Figure 4: **Average of Effects of Migration on the Share of Evangelicals among Native Populations**



The light-colored bars represent the average change in Evangelical share across municipalities, weighted by each municipality’s 1980 population, in 1980–1991, 1991–2000, and 2000–2010. The dark-colored bars depict the counterfactual scenario with zero migration rates. Formally, the counterfactual estimates are calculated as $\Delta s_d^{cf} = \Delta s_d - \hat{\beta} m_d$, where Δs_d denotes the change in the share of Evangelicals in municipality d during a given period, $\hat{\beta}$ is the (SSIV) point estimate from Table 2 for the corresponding period, and m_d represents the migration rate in municipality d . These values are then averaged over municipalities, weighted by their 1980 populations.

Finally, to provide a clearer perspective on the magnitude of the estimated effects, Figure 4 illustrates the variation in the share of Evangelicals across the three periods—1980–1991, 1991–2000, and 2000–2010—under both baseline and counterfactual scenarios in which migration rates are set to zero in all municipalities. The light-colored bars represent the average change in the Evangelical share across municipalities (i.e., the outcome variable in the regressions for Evangelicals), weighted by each municipality’s 1980 population. The dark-colored bars show the same averages in a counterfactual scenario where migration rates are assumed to be zero. Formally, the counterfactual estimates are computed as $\Delta s_d^{cf} = \Delta s_d - \hat{\beta} m_d$, where

Δs_d denotes the change in the share of Evangelicals in municipality d during a given period, $\hat{\beta}$ is the corresponding (SSIV) point estimate from Table 2, and m_d represents the migration rate in municipality d in that period. These values are then averaged across municipalities, weighted by their 1980 populations.

Back-of-the-envelope calculations based on Figure 4 indicate that migration has significantly influenced the growth of the Evangelical population in Brazil. For example, between 1991 and 2000, the average increase in the share of Evangelicals was approximately 5.5 percentage points in the baseline scenario, compared to 3.5 percentage points in a counterfactual scenario without migration. Cumulatively, over the three periods analyzed, the difference between these scenarios amounts to 3.7 percentage points, suggesting that the effects we are estimating are economically relevant.

3.3 Robustness

In this section, we conduct a series of robustness checks to evaluate the reliability of our main results.

First, concerns regarding the potential correlation between the instrument and unobserved factors affecting religious conversion appear to be limited in our context. Including controls for regional economic and demographic trends has minimal impact on the estimates, as shown in Tables B2 and B3—compare columns 1 (without controls) and 2 (with controls). In particular, the results remain stable when controlling for differential trends in key demographic characteristics, household income per capita, total population, and the lagged share of each religion.

Second, one might worry that destination food prices are correlated with commodity price shocks in the areas of origin. If poverty influences conversion, this correlation could threaten the validity of our identification. To address this concern, column 3 of both tables includes destination price shocks alongside the baseline. The estimates remain stable, with only minor changes observed for both Catholics and Evangelicals.

Third, we consider the possibility that price shocks in origin areas might affect destination economies through trade channels, as noted by Corbi et al. (2025), Imbert and Ulyssea (2022), and Kleemans and Magruder (2018). For example, destination municipalities may

industrialize or strengthen commercial ties with origin regions. To account for this, we construct a measure of price shocks at the destination, weighted by the inverse distance to each origin. The rationale is that trade flows are generally stronger between municipalities that are closer to each other. Column 4 demonstrates that controlling for this measure does not alter our results.

Fourth, we investigate whether variations in religious composition across states affect our results. A comparison of columns 2 and 5, which includes state fixed effects, shows that our estimates are robust against this concern.

Fifth, we test the validity of the common pre-trend assumption. If municipalities exposed to migration shocks exhibit diverging pre-shock religious trends, our estimates might reflect those differences rather than a causal effect. Following Corbi et al. (2025) and Imbert and Ulyssea (2022), we incorporate the lagged dependent variable. Results in column 6 confirm that our main findings hold after accounting for these pre-existing trends.

Finally, we consider the persistence of migration. If migration flows are highly persistent, our instrument may reflect prior waves rather than current shocks (Jaeger et al., 2018). In column 7, we follow Imbert and Ulyssea (2022) and control for lagged migration. Our results remain stable across all specifications, and our baseline already accounts for population, which helps address long-term migration effects.

4 Mechanisms

An important finding from the previous section is that migration has affected the religious landscape in Brazil in a very particular way: it led to a substantial decline in the share of Catholics and a corresponding rise in the share of Evangelicals *among native, non-migrant, populations of Brazilian cities*. However, the economic mechanisms driving this shift, which are embedded in the β estimate, remain unclear. This section aims to explore and identify plausible underlying economic channels that may explain the asymmetric effects of migration on the shares of Catholics and Evangelicals. As previously anticipated, two key channels frequently highlighted in the literature may underlie this pattern: Changes in labor market conditions and the expansion of religious infrastructure. Both channels link migration to

shifts in the religious composition of the Brazilian population.

Migration, labor markets, and religious choices. The demand-side explanation for the impact of migration waves on the religious composition of native populations relates to two strands of literature. First, recent studies emphasize the broader adverse macroeconomic effects of migration shocks on local labor markets in Brazil (Imbert and Ulyssea, 2022; Corbi et al., 2025). Second, a separate body of research attributes the rise of Evangelicals to adverse economic conditions (Costa et al., 2019), arguing that economic insecurity increases individuals’ reliance on institutions that provide material, emotional, and social support. In this context, Evangelical churches have been particularly effective in addressing economic and social hardships (Chesnut, 1997; Costa et al., 2019). Through their doctrines, practices, and support networks—reminiscent of the concept of social capital described by Putnam (2000)—Evangelicalism offers empowerment and assistance to individuals often left without institutional support from the Brazilian state.

While other religious groups, particularly the Catholic Church, also seek to create inclusive spaces for the poor, Pentecostal churches have been notably more effective in doing so from a demand-side perspective (Costa et al., 2019; Chesnut, 1997). Critics argue that Catholic priests tend to be more distant and less responsive to the immediate needs of their congregations. Although the Catholic Church in Brazil has made efforts to popularize its practices and strengthen community engagement (Scott, 1989), Catholic rituals are often perceived as more traditional, less dynamic, and more challenging to grasp. Moreover, Catholic responses to poverty typically emphasize structural social change (Martin, 1990), whereas Pentecostal churches provide more immediate and tangible forms of support.

These insights suggest that migration-induced economic shocks may have played a crucial role in reshaping religious affiliation among native populations by increasing reliance on Evangelical churches as providers of social and economic support. If economic insecurity leads individuals to seek institutions that offer immediate relief, then migration’s effect on labor markets could be a key mechanism explaining the religious transition observed in Brazil. To evaluate this channel, we first establish the link between migration and income shocks in local labor markets. In the next section, we return to this question using a structural model

of religious demand to assess how income variation shapes religious choices. Specifically, we estimate a model similar to equation (1), using variation in (log) average labor income of native workers at the municipality level as the outcome variable. Migration is instrumented using the same SSIV approach discussed in the previous section. The results, presented in Table 3, provide evidence on the labor market consequences of migration. All coefficients and standard deviations (in brackets) in Table 3 were estimated from the origin-level specification discussed earlier. Standard deviations are clustered at the origin municipality level. Regressions include the same baseline covariates as in Table 2.

We present results only for the 1991–2000 and 2000–2010 periods due to data limitations in the 1980 Census, including high non-response rates for labor market variables and inconsistencies with later Census waves. Columns 1–3 report results using average individual labor income as the outcome variable, while column 4 examines household income per capita. Column 2 focuses on formal workers, and column 3 presents results for informal workers. Panels A and C display OLS estimates for 1991–2000 and 2000–2010, respectively, while Panels B and D show the corresponding SSIV estimates.

Overall, the results indicate that migration shocks depress both average labor income and household income per capita, with especially pronounced effects among informal workers. These findings are consistent with previous research—such as Corbi et al. (2025) and Imbert and Ulyssea (2022)—which also draw on Brazilian Census data and employ similar identification strategies to analyze the labor market consequences of migration. Based on the same approach illustrated in Figure 4, back-of-the-envelope calculations suggest that, in the absence of migration flows, average wages across years and municipalities would have been 4.1% higher in 2000 and 6.3% in 2010.

We conducted several robustness checks to validate these results. Table B4 in the Appendix presents robustness tests for the overall earnings of native workers. Given concerns about pre-trends, persistence of migration flows, and correlations between price shocks and earnings, we apply the same robustness tests as in the previous section. To control for potential pre-existing trends, we follow Corbi et al. (2025) and Imbert and Ulyssea (2022) by including the lagged dependent variable (column 6). We control for lagged migration rates to address concerns about persistent migration flows (column 7). Additionally, we account

Table 3: **Effects of Migration on Earnings for Native Workers**

	(1)	(2)	(3)	(4)
	Overall	Formal	Informal	Family income
Panel A: OLS (Δ 1991-2000)				
In-migration rate	-0.019 (0.026)	0.009 (0.032)	-0.030 (0.031)	-0.018 (0.027)
Observations	3,919	3,919	3,919	3,919
Panel B: SSIV (Δ 1991-2000)				
In-migration rate	-0.401*** (0.092)	-0.217** (0.088)	-0.360*** (0.101)	-0.437*** (0.118)
First Stage F Stat	192.2	192.2	192.2	192.2
Observations	3,989	3,989	3,989	3,989
Panel C: OLS (Δ 2000-2010)				
In-migration rate	-0.012 (0.020)	-0.055** (0.026)	0.048** (0.020)	-0.004 (0.020)
Observations	3,919	3,919	3,919	3,919
Panel D: SSIV (Δ 2000-2010)				
In-migration rate	-0.540*** (0.199)	-0.435*** (0.149)	-0.618** (0.263)	-0.579*** (0.208)
First Stage F Stat	70.9	70.9	70.9	70.9
Observations	3,989	3,989	3,989	3,989
Baseline covariates	✓	✓	✓	✓

This table shows municipality of destination-level OLS and origin-level SSIV coefficients on the 5-year accumulated changes in the municipality-level (log) earnings for native workers in destination municipalities. The SSIV is the inflow of migrants predicted by price shocks at the origin municipality. In columns 1-3, the dependent variable is the change in the destination municipality's average labor earnings, while in column 4 is the change in the municipality's average household income per capita. In Panels A and B, the dependent variable is the change in the outcome between 1991 and 2000, while in Panels C and D, it is the change between 2000 and 2010. All regressions include baseline lagged covariates (share of population less than 18 years old; share of non-white; share of female; share with complete high school or higher; share of married people in the population; household income per capita, and the total population) and are weighted by the total population in 1980. Standard deviations are clustered at the origin municipality level. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

for possible correlations between destination food prices and origin price shocks (column 3). Across all specifications, our results remain stable.¹⁶ Including additional controls (compare columns 1 and 2) and state fixed effects (column 5) has little impact on our estimates. Controlling for trade linkages between destination and origin municipalities (column 4) also has

¹⁶The only exception is when lagged migration is included (column 7): the estimate for 2000–2010 becomes statistically insignificant, though the point estimate remains negative and sizable. The effect for 1991–2000 remains statistically significant.

no consequences for our findings.

Migration and supply expansion. On the supply side, it is likely that churches may have reacted to massive population shifts by expanding infrastructure. Evangelical churches, in particular, have demonstrated an unparalleled ability to expand. Importantly, various evidence seems to indicate that they have expanded particularly faster in areas that received larger migration influxes. This expansion stands in sharp contrast to the Catholic Church, which often relies on centralized, hierarchical structures and slower institutional processes, making it less adaptable to the rapidly changing environment brought about by internal migration (Corbi and Sanches, 2024).

Figure 5 provides descriptive evidence that corroborates these points. First, Panel (a) shows that while the number of Catholic temples doubled between 1980 and 2010, the number of active Evangelical temples increased almost sevenfold during the same period. In addition, Panel (b) reveals that this massive expansion of Evangelical temples was concentrated in cities with higher migration rates, perhaps suggesting that the Evangelical supply-side actively responded to migration shocks.

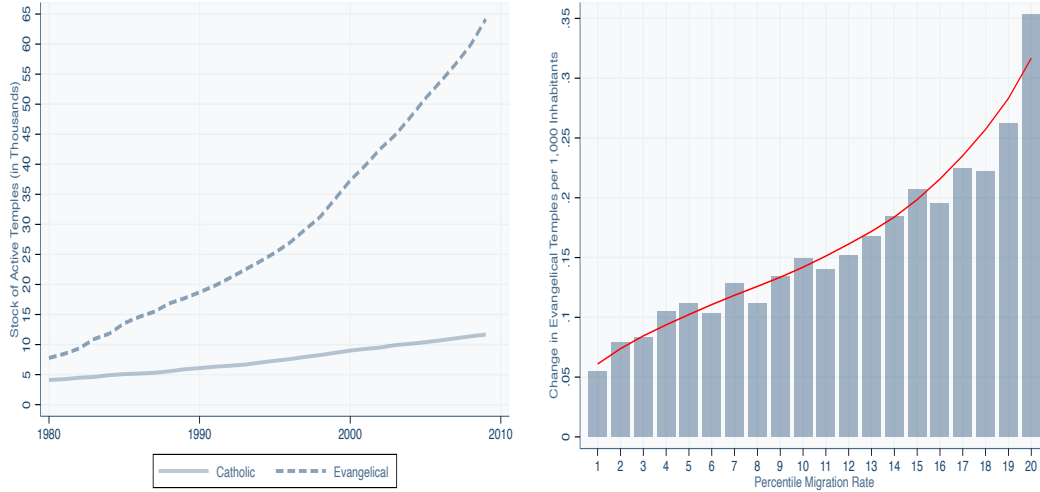
More formally, we analyze how Evangelical and Catholic churches respond to migration shocks by estimating model (1), using the change in the number of temples normalized by the population in the baseline year in each municipality as the outcome variable. Put simply, we seek to determine whether Catholic and Evangelical churches expanded their religious infrastructure in response to migrant inflows. In our context, such a response would support the following supply-side mechanism: the arrival of migrant waves prompted churches to increase the supply of temples, which, in turn, may have also accommodated new native adherents.¹⁷

Table 4 provides evidence of the positive effect of migration on temple supply. Columns 1 and 2 present the OLS and SSIV estimates of migration on the number of Catholic temples, while columns 3 and 4 report the corresponding effects on Evangelical temples. Panels A, B, and C display these estimates for the 1980–1991, 1990–2000, and 2000–2010 periods, respectively. Coefficients and standard errors are estimated from the origin-level specification;

¹⁷Migration may have affected temple supply through additional channels beyond direct responses to demographic growth. We discuss these alternative mechanisms later in this section and the next.

standard errors are clustered at the origin municipality level. All regressions control for the same set of baseline covariates as those reported in Table 2 and are weighted by the total population in 1980.

Figure 5: **Supply Side Behavior of Catholics and Evangelicals**



(a) Stock of temples (1980-2010)

(b) Migration and Pentecostal temples

Panel (a) shows the stock of active temples of the Catholic Church and Evangelical denominations in Brazil from 1980 until 2010. Data source: Internal Revenue Service. Panel (b) shows the relationship between (in)migration rates and the per capita growth of Evangelical temples across Brazilian municipalities between 1980-1991, 1991-2000, and 2000-2010. The x-axis represents percentiles of migration rates, ranging from the 1st (municipalities with the lowest migration rates) to the 20th (municipalities with the highest migration rates). The bars represent the average yearly variation (in units per 1000 inhabitants) of Pentecostal temples within each percentile, while the line represents the LOWESS smoothed series. Data source: Brazilian Census from 1980-2010 and Internal Revenue Service.

From these results, we highlight the following key points. First, the effect of migration on the construction of Catholic temples is positive and statistically significant only in the first period. However, even in this case, the magnitude of the effect is substantially smaller than the corresponding estimate for Evangelical churches. Second, migration appears to have consistently contributed to the expansion of Evangelical temples across all periods. As in the previous section, SSIV estimates for Evangelical churches (Catholic Church) are generally larger (smaller) than the corresponding OLS estimates. This pattern may be driven by the tendency of migrants to settle in economically stronger regions, where the appeal of Evangelical churches may be relatively weaker and that of the Catholic Church is more pronounced.

Table 4: **Effects of Migration on the Number of Temples of each Religious Group**

	Dep. Var.: Δ number of temples per capita			
	Catholics		Evangelicals	
	(1)	(2)	(3)	(4)
	OLS	SSIV	OLS	SSIV
<u>Panel A: Δ 1980-1991</u>				
In-migration rate	0.096*** (0.020)	0.079*** (0.016)	0.539*** (0.079)	0.883*** (0.151)
First Stage F Stat		185.9		185.9
Observations	3,919	3,989	3,919	3,989
<u>Panel B: Δ 1991-2000</u>				
In-migration rate	0.031*** (0.011)	-0.021 (0.015)	0.309*** (0.044)	0.685*** (0.128)
First Stage F Stat		192.2		192.2
Observations	3,919	3,989	3,919	3,989
<u>Panel C: Δ 2000-2010</u>				
In-migration rate	0.020*** (0.005)	-0.006 (0.014)	0.164*** (0.030)	0.651** (0.255)
First Stage F Stat		70.9		70.9
Observations	3,919	3,989	3,919	3,989
Baseline covariates	✓	✓	✓	✓

This table presents municipality of destination-level OLS and municipality of origin-level SSIV estimates of the effect of migration on the number of Catholic and Evangelical temples per inhabitant. Columns 1 and 2 show estimates for Catholic temples, while Columns 3 and 4 report estimates for Evangelical temples. The SSIV uses migration inflows predicted by price shocks at the origin. Panels A, B, and C display the 1980-1991, 1991-2000, and 2000-2010 results, respectively. All regressions include controls for lagged baseline covariates (share of population under 18 years old; share of non-white; share of female; share with high school education or higher; share of married individuals; household income per capita, and total population). Regressions are weighted by the total population in 1991. Standard errors are clustered at the origin municipality level. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

The faster and more intense response of Evangelical churches to demand shocks, compared to the Catholic Church, reflects fundamental differences in their business models, as highlighted by Corbi and Sanches (2024). While the Catholic Church faces high entry costs—requiring significant investments in building temples and lengthy training periods for new priests—Evangelical churches operate with much lower costs, typically renting spaces and training pastors quickly or with minimal formal requirements. The results in Table 4 thus appear consistent with these structural differences, reinforcing the idea that lower entry

costs and greater flexibility in Evangelical church expansion drive their faster and more pronounced response to migration-driven demand shocks, while the Catholic Church’s higher fixed costs limit its responsiveness (Corbi and Sanches, 2024).

We also compute back-of-the-envelope calculations to illustrate the impact of migration on the number of Evangelical temples per capita over the periods 1980–1991, 1991–2000, and 2000–2010. These calculations follow the same methodology used in Figure 4. The results suggest that, in the absence of migration, the number of temples per 1,000 inhabitants would be approximately 27% lower on average across years and municipalities—corresponding to declines of 38% in 1991, 25% in 2000, and 20% in 2010. These magnitudes highlight the relevance of this channel in accounting for the findings discussed in Section 3.

There are at least two mechanisms through which migration may have produced this substantial effect. First, by depressing local wages, migration may have shifted religious demand toward Evangelicalism; in equilibrium, churches would respond by increasing temple supply. Second, by increasing the local population, migration expanded the religious market, creating incentives to open more temples across the territory, serving both migrants and non-migrants alike. The structural model developed in the next section allows us to assess the relative importance of these two explanations.

To conclude the section, Tables B5 and B6 present robustness checks for the results reported in Table 4. These checks confirm that our main conclusions remain unchanged when we control for origin–destination correlations in price shocks (column 3), trade linkages (column 4), and a broad set of economic and demographic controls, including population composition, income levels, and state fixed effects (columns 1, 2, and 5). The main deviation arises when we include pre-trends (column 6) and lagged migration (column 7) as controls. In these cases, the coefficient for Evangelicals (only) in the 2000–2010 period becomes less precise, although the point estimates remain positive and of meaningful magnitude.

5 Quantifying the Mechanisms

To quantify the importance of the supply- and demand-side channels analyzed above, this section develops and estimates a structural model of religious demand. We adopt a discrete

choice framework in the spirit of Berry et al. (2004), in which individuals’ preferences over religious options depend on two key factors:¹⁸ their income—since the appeal of different religions may vary across income levels—and the local stock of temples for each religion, which captures the role of infrastructure availability in shaping religious choice. We estimate the model and use it to simulate counterfactual scenarios that isolate the contribution of each mechanism. While these exercises are not causal, they offer quantitative insight into how much of the observed rise in Evangelical affiliation could plausibly be attributed to each channel. More broadly, the model allows us to connect migration-induced changes in income and infrastructure to shifts in religious demand, thereby clarifying the specific channels through which migration contributed to Brazil’s religious transformation.

5.1 Framework

The demand for religion focuses on how individuals substitute between the Catholic Church, Evangelical churches, and the outside option—i.e., no religion—depending on individuals’ and religions’ characteristics.¹⁹ To formalize this, we consider an individual i residing in municipality d in a given period.²⁰ The individual’s objective is to choose a religion $j \in \chi = \{0, C, E\}$ that maximizes her utility, where 0 represents the outside option, C the Catholic Church, and E an Evangelical denomination. The utility of individual i from choosing religion $j \neq 0$ in municipality d is given by:

$$u_{ijd} = \pi_j + \gamma_j \alpha_{id} + \delta_{jd} + \varepsilon_{ijd}, \quad (7)$$

where u_{ijd} is the total utility derived from religion j . The first term, π_j , represents the baseline preference for religion j that is common across individuals and municipalities. The second term, $\gamma_j \alpha_{id}$, introduces individual-level heterogeneity in preferences. Here, α_{id} cap-

¹⁸See also Corbi et al. (2022) for an application of this approach to religious demand in the U.S.

¹⁹To keep the model estimation tractable, we exclude other religions. This group’s share is relatively small (on average, 1% of the population) and remains fairly stable over time. Moreover, the primary religious transition we analyze involves shifts from the Catholic Church to Evangelical denominations, which our model flexibly accommodates.

²⁰For simplicity, we omit time notation from our models.

tures how the taste for attributes of religion j , summarized by γ_j , varies with individual characteristics. The third term, δ_{jd} , captures the influence of religion-municipality specific factors, such as the availability of religious infrastructure in different markets (e.g., the number of temples), on the utility individuals derive from religion j . Finally, ε_{ijd} is an Extreme Value Type I error term, assumed to be iid across individuals, religious options, and municipalities. This assumption is standard in models of demand for differentiated products; see, for example, Berry and Haile (2021) for a complete overview of this literature.

The heterogeneous component of preferences, α_{id} , is modeled as:

$$\alpha_{id} = \alpha_0 + \alpha_1 w_{id}(m_d) + \sum_{l=1}^L \alpha_{2l} x_{lid}, \quad (8)$$

where α_0 is a baseline preference common to all individuals; the term $w_{id}(m_d)$ represents income, and the coefficient α_1 measures how preferences for religion j vary with income. We write it as a function of migration rates to municipality d , m_d , to highlight the fact that income w_{id} is affected by labor market fluctuations arising from migration shocks, as suggested by the estimates reported in Table 3. The summation $\sum_{l=1}^L \alpha_{2l} x_{lid}$ incorporates the effects of demographic characteristics x_{lid} , such as age, education, race, and gender, with α_{2l} capturing their respective impacts on preferences.²¹ This specification allows preferences for religion to vary systematically with individuals' demographic characteristics, as well as with income changes induced by labor market shocks.

The component δ_{jd} captures how the presence of religious infrastructure in municipality d and other religion-municipality specific factors influence the utility of religion j . Specifically, we model δ_{jd} as:

$$\delta_{jd} = \delta_{0j} + \delta_{1j} N_{jd}(m_d) + \xi_{jd}, \quad (9)$$

²¹In principle, we assume that education is not influenced by migration. Education is costly and unlikely to respond to labor market shocks in the short run. Nevertheless, as we argue below, our objective is to evaluate the effects of migration on the demand for religion through the two channels highlighted in the previous section. The exercises we present remain informative regardless of the existence of other mechanisms through which migration shocks may influence the demand for religion.

where δ_{0j} represents a religion specific constant j , and $N_{jd}(m_d)$ denotes the number of temples of religion j in municipality d normalized by its population in 1980. The coefficient δ_{1j} captures the impact of religious infrastructure on utility. A greater availability of temples reduces the cost of accessing religious services and may enhance the visibility or perceived legitimacy of a religion within the community. The term ξ_{jd} accounts for unobserved, municipality-specific factors affecting the attractiveness of religion j .²² Churches may respond to demand shocks induced by migration waves by adjusting $N_{jd}(m_d)$ —as the results in Table 4 strongly suggest. We write it as a function of m_d to emphasize this channel. Through this process, we aim at inferring the sensitivity of the demand for religion to supply-side responses to migration shocks.

Substituting equations (8) and (9) into equation (7), we express the utility of individual i from choosing religion j in municipality d as:

$$u_{ijd} = \tilde{\delta}_{jd}(m_d) + \tilde{\gamma}_j w_{id}(m_d) + \sum_{l=1}^L \tilde{\alpha}_{jl} x_{lid} + \varepsilon_{ijd}, \quad (10)$$

where $\tilde{\gamma}_j = \gamma_j \alpha_1$ and $\tilde{\alpha}_{jl} = \gamma_j \alpha_{2l}$ represent, respectively, the effect of income and demographic variables on the utility individuals derive from Church j . The term $\tilde{\delta}_{jd}(m_d)$ is defined as $\tilde{\delta}_{jd}(m_d) = \tilde{\delta}_{0j} + \delta_{1j} N_{jd}(m_d) + \xi_{jd}$ where, $\tilde{\delta}_{0j} = \pi_j + \gamma_j \alpha_0 + \delta_{j0}$. Therefore, $\tilde{\delta}_{jd}(m_d)$ absorbs Church and church-municipality effects, which include the number of temples and any unobserved municipality-level shock specific to religion j , on the utility for religion j .²³

The individual's choice set also includes an outside option, denoted by $j = 0$, which represents the non-religious, or secular, option. Following convention in the literature, we normalize the utility of the outside option as follows:

$$u_{i0d} = \varepsilon_{i0d}, \quad (11)$$

where ε_{i0d} is an Extreme Value Type I shock, assumed to be iid across individuals i and

²²Again, it is likely that this term also depends on migration. However, as we argued earlier, this does not affect the conclusions presented in this section.

²³Note that the model does not explicitly account for inertia or for switching costs in religious choices as in Corbi et al. (2022). However, this is unlikely to be a major concern in our setting, as most individuals were initially Catholic. Given this initial condition, religious heterogeneity was relatively low, and inertia in religious choices does not pose a significant threat to our analysis.

municipalities d . This assumption is innocuous in the sense that it only changes the interpretation of the estimates of the parameters in equation (10) from “levels” to “relative to the outside option” without influencing our counterfactuals.

Given these assumptions, if individuals choose the alternative $j \in \chi$ that maximizes their utility, then the demand for religion j of an individual residing in a municipality d can be characterized as:

$$S_{jd}(m_d, \mathbf{x}_{id}) = \begin{cases} \frac{\exp(\tilde{\delta}_{jd}(m_d) + \tilde{\gamma}_j w_{id}(m_d) + \sum_{l=1}^L \tilde{\alpha}_{jl} x_{lid})}{1 + \sum_{k \in \chi \setminus 0} \exp(\tilde{\delta}_{kd}(m_d) + \tilde{\gamma}_k w_{id}(m_d) + \sum_{l=1}^L \tilde{\alpha}_{kl} x_{lid})} & \text{if } j \in \chi \setminus 0, \\ \frac{1}{1 + \sum_{k \in \chi \setminus 0} \exp(\tilde{\delta}_{kd}(m_d) + \tilde{\gamma}_k w_{id}(m_d) + \sum_{l=1}^L \tilde{\alpha}_{kl} x_{lid})} & \text{if } j = 0. \end{cases} \quad (12)$$

In equation (12) $\mathbf{x}_{id}$ represents the vector of individuals demographic characteristics. The aggregate demand for religion j in a municipality d is obtained by integrating equation (12) over the joint distribution of income and individual characteristics in each municipality:

$$\bar{S}_{jd}(m_d) = \begin{cases} \int \frac{\exp(\tilde{\delta}_{jd}(m_d) + \tilde{\gamma}_j w(m_d) + \sum_{l=1}^L \tilde{\alpha}_{jl} \mathbf{x}_l)}{1 + \sum_{k \in \chi \setminus 0} \exp(\tilde{\delta}_{kd}(m_d) + \tilde{\gamma}_k w(m_d) + \sum_{l=1}^L \tilde{\alpha}_{kl} \mathbf{x}_l)} dF_d(w(m_d), \mathbf{x}) & \text{if } j \in \chi \setminus 0, \\ \int \frac{1}{1 + \sum_{k \in \chi \setminus 0} \exp(\tilde{\delta}_{kd}(m_d) + \tilde{\gamma}_k w(m_d) + \sum_{l=1}^L \tilde{\alpha}_{kl} \mathbf{x}_l)} dF_d(w(m_d), \mathbf{x}) & \text{if } j = 0. \end{cases} \quad (13)$$

The joint cumulative distribution function $F_d(w(m_d), \mathbf{x})$ captures the distribution of income $w(m_d)$ and the demographic vector $\mathbf{x}$ within municipality d . By integrating over this distribution, the aggregate demand $\bar{S}_{jd}$ accounts for the heterogeneity in individual characteristics across municipalities.

5.2 Estimation and Results

We now outline the estimation of the structural model parameters, followed by a discussion of the baseline results.

Estimation Estimation is conducted in two steps. In the first step, we estimate equation (12)—i.e., the parameters $\tilde{\delta}_{jd}$, $\tilde{\gamma}_j$, and $\tilde{\alpha}_{jd}$ for all $j \neq 0$, l , and d —using Maximum Likelihood

with individual-level data from the Brazilian Census.

Step 1 is computationally intensive due to the large number of observations and municipality-religion-specific effects, $\tilde{\delta}_{jd}$, that need to be estimated. To make the estimation feasible, we use a sample comprising 4% of the total microdata from the Brazilian Censuses of 1991, 2000, and 2010.²⁴ Again, we do not use the 1980 sample due to the problems with the information on household income in that Census wave. Additionally, we excluded individuals younger than 16 years from the sample, as younger individuals are less likely to be actively making independent religious choices; instead, their religious affiliations may reflect those of their parents or caregivers. As before, the model is estimated separately for each year.

To estimate $\tilde{\delta}_{jd}$ within each municipality, a sufficiently large number of observations is necessary. Municipalities with fewer than 450 observations in the sample were excluded, as reducing this threshold often caused the models to fail to converge. After these adjustments, the sample for 1991 includes 188,978 observations in 181 municipalities; for 2000, it includes 257,335 observations in 249 municipalities; and for 2010, it includes 245,113 individuals in 305 municipalities.²⁵ As a result, our sample focuses on the largest Brazilian municipalities, which is where the religious transition was concentrated.

The demand model estimation proceeds to Step 2, where we use the $\tilde{\delta}_{jd}$ to estimate $\tilde{\delta}_{jd}(m_d) = \tilde{\delta}_{0j} + \delta_{1j}N_{jd}(m_d) + \xi_{jd}$. The main challenge is the potential endogeneity of N_{jd} , which may be correlated with unobserved market-level factors, ξ_{jd} . Following Berry et al. (2004), separating demand estimation into two steps helps mitigate the impact of this issue on our results by ensuring that the estimates of $\tilde{\gamma}_j$ remain unaffected.

To address this concern, our baseline second-stage regression pools the three Census waves (1991, 2000, and 2010), interacts temple density with Census wave dummies, and includes both municipality-religion and year-religion fixed effects. Specifically, for each group j we estimate separately $\tilde{\delta}_{jdt} = \sum_t \delta_{1jt} N_{jdt} d_t + \psi_{jd} + \phi_{jt} + \varepsilon_{jdt}$, where d_t s are dummies for the 1991, 2000 and 2010 Census waves, ψ_{jd} are municipality-religion fixed effects, ϕ_{jt} are

²⁴Using larger samples is infeasible because the computational time required to estimate the model grows exponentially with the number of observations. The selected sample is sufficiently large to provide robust estimates while keeping computation time manageable.

²⁵The total number of observations in the 1991 sample is substantially smaller than in the other years. To account for these differences, all estimates will be weighted using individual (and municipality) Census weights specific to each year.

year–religion fixed effects and ε_{jdt} are transitory local demand shocks. Municipality fixed effects account for long-run differences in religious preferences driven by geography, history, and culture, while year fixed effects capture nationwide religious trends. Under this specification, identification of δ_{1j} relies on the assumption that church investments in infrastructure are uncorrelated with short-term fluctuations in unobserved church–municipality demand shifters around their long-term average, a premise we substantiate with additional evidence later in this section.²⁶

Next, we show our baseline estimates and then turn to counterfactual simulations that quantify the relative contributions of income and infrastructure channels to the migration-driven rise in Evangelical affiliation. In the concluding subsection, we undertake a comprehensive set of robustness exercises and bring additional evidence that lends further support to our identifying assumptions and to our main findings.

Baseline results. Tables C7 and C8 in the Appendix report our key estimates. We estimate two versions of the model. The first includes demographic controls—education, age, and gender—as specified in the covariates x_{lid} in equation (8); the second omits these controls.²⁷

For conciseness, Table C7 reports only the key parameter $\tilde{\gamma}_j$ for Catholics and Evangelicals. Standard errors, clustered at the municipality level, are shown in parentheses. Panel A presents estimates from the model with demographic controls; Panel B omits them. Two main patterns emerge. First, across all specifications and years, $\tilde{\gamma}_j$ for Evangelicals is consistently negative and highly significant, indicating that lower income is associated with a higher probability of choosing an Evangelical church. This relationship is robust to the inclusion of demographic controls. Second, estimates for the Catholic Church show no sys-

²⁶Although investments in temples (N_{jd}) may respond in part to short-term shifts in demand, they are largely shaped by persistent structural conditions, such as local historical, cultural, and economic factors. Building a temple requires planning, fundraising, land acquisition, and regulatory procedures, especially in the case of the Catholic Church, where infrastructure is more capital-intensive and hierarchical. Even among Evangelicals—who are organizationally more flexible—church entry requires coordination and capital, introducing some inertia. Unlike prices in standard demand estimation models, which adjust quickly to transitory shocks, religious infrastructure tends to evolve more gradually, responding primarily to long-term shifts in market conditions.

²⁷Accordingly, Step 1 yields two sets of $\tilde{\delta}_{jd}$ estimates—one from the model with demographic controls and one without. In Step 2, we run separate regressions using each set of $\tilde{\delta}_{jd}$ values.

tematic pattern and are generally not statistically significant, suggesting that income does not predict Catholic affiliation in a consistent way.

Table C8 reports the second-stage estimates of δ_{1j} , using both Pooled OLS and Two-Way Fixed Effects (TWFE) models, for versions with and without demographic controls in the first stage. Standard errors are clustered at the municipality level. In the Pooled OLS models, Evangelical temple supply is positively associated with demand in 1991 and 2000, though the estimates are imprecise and generally not statistically significant; the Catholic coefficients are larger and significant. In contrast, the TWFE models reveal a different pattern: the Catholic coefficients shrink and lose significance, while the Evangelical coefficients become larger and highly significant.

One notable feature of Table C8 is that the positive association between Evangelical temple density and the estimated demand shifter emerges clearly only after controlling for fixed effects. For the Catholic Church, the opposite occurs: a strong positive association in the OLS specification weakens substantially once fixed effects are introduced. This pattern is informative in evaluating potential endogeneity concerns. If unobserved demand shocks correlated with temple supply were driving the estimates, we would expect the coefficients to increase—rather than diminish—after controlling for fixed effects. Instead, we observe the reverse for Catholics and a pronounced strengthening for Evangelicals once long-run heterogeneity is absorbed, suggesting that omitted variable bias is unlikely to be the source of our main results. More broadly, this pattern mirrors the results in Tables 2 and 4, where SSIV estimates are consistently larger than OLS for Evangelicals, and the opposite holds for the Catholic Church. We return to this point later in the section and provide further evidence supporting the identifying assumptions behind our estimates.

In any case, these findings offer initial evidence that religious demand responds through both income and infrastructure channels. The negative income gradient among Evangelicals suggests that lower income is associated with a higher likelihood of Evangelical affiliation, in line with prior work (Costa et al., 2019; Chesnut, 1997). At the same time, we find that Evangelical affiliation is strongly associated with local temple availability. Combined with the results from the previous section—showing that migration led to income declines and a sharp rise in Evangelical temple supply—these preliminary results support the view that

migration altered the local conditions under which individuals make religious choices through demand and supply channels.

5.3 Counterfactuals

While the estimates in Tables C7 and C8 show that religious affiliation responds to both income and infrastructure, they do not directly quantify the relative importance of each channel. This section presents a set of counterfactual exercises designed to assess how much of the effect of migration on Evangelical growth—documented earlier in the paper—can be attributed to changes in income versus changes in religious infrastructure.

Our approach is as follows. Using the demand model, we first simulate the effect of migration on the share of Evangelicals, under the assumption that migration influenced religious affiliation exclusively through changes in income and Evangelical temple supply. Mathematically, this corresponds to computing the following difference:

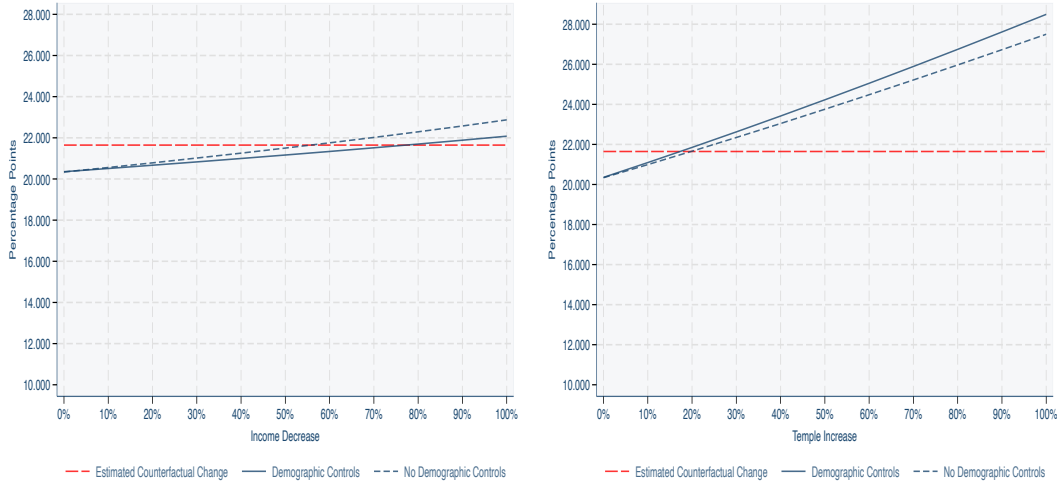
$$\Delta \bar{S}_E = \frac{1}{\#(i, d, t)} \sum_{i, d, t} [S_{Edt}(w_{idt}, N_{Cdt}, N_{Edt}) - S_{Edt}(w_{idt}^o, N_{Cdt}^o, N_{Edt}^o)],$$

where $S_{Edt}(w_{idt}, N_{Cdt}, N_{Edt})$ denotes the share of Evangelicals in municipality d and year t predicted by the model using observed wages and temple stocks, and $S_{Edt}(w_{idt}^o, N_{Cdt}^o, N_{Edt}^o)$ is the predicted share under counterfactual (pre-migration) income, w_{idt}^o , and temple stocks, N_{Cdt}^o and N_{Edt}^o . The difference $\Delta \bar{S}_E$ represents the structural model analogue of the migration effect estimated in Section 3, under the assumption that migration affects religious affiliation only through the income and infrastructure channels.

We then scale pre-migration income, w_{idt}^o , downward uniformly, keeping temple stocks fixed at their pre-migration levels, until $\Delta \bar{S}_E = 0$. This yields the decline in income that would be required—on its own—to reproduce the effect of migration on the Evangelical share. We repeat the same procedure for pre-migration Evangelical temple supply, N_{Edt}^o , holding income fixed at pre-migration levels, to compute the change in infrastructure that would be necessary, by itself, to generate the same effect. Comparing these required changes with the actual migration-induced change obtained in the previous section allows us to assess which mechanism more plausibly accounts for the observed shift in affiliation. If reproducing the

rise in Evangelical affiliation requires changes in income or infrastructure that far exceed the migration-induced changes estimated in the previous section, then those mechanisms are less likely to account for the observed effect. Conversely, if the required and estimated changes are of similar magnitude, the mechanism becomes a more plausible explanation.²⁸

Figure 6: **Changes in Income and Temple Supply Required to Rationalize the Observed Effects of Migration on the Share of Evangelicals**



(a) Income Counterfactual

(b) Temple Counterfactual

Panel (a) shows the counterfactual Evangelical share under proportional income reductions, with $\varrho \in [0, 1]$, where $\varrho = 0$ denotes the baseline and $\varrho = 1$ eliminates income. Temple stock is fixed at its no-migration level. Panel (b) simulates proportional increases in Evangelical temples with income and Catholic temples held fixed at pre-migration levels. The dashed (solid) line refers to models estimated without (with) demographic controls. The horizontal line denotes the baseline Evangelical share $\bar{S}_E$, and intersection points indicate the required change under each mechanism to rationalize the observed Evangelical share.

The results of this exercise are presented in Figure 6. To construct each panel, we take the estimated demand model and evaluate it at pre-migration levels of income and temple supply for each individual in the sample. These pre-migration values— w_{idt}^o , N_{Cdt}^o , and N_{Edt}^o —are derived using the estimated effects of migration on income and infrastructure reported in Tables 3 and 4. Further details on their construction are provided in the Appendix. The upward-sloping curves in Figure 7(a) plot the average of $S_{Edt}((1 - \varrho_w)w_{idt}^o, N_{Cdt}^o, N_{Edt}^o)$

²⁸One limitation of this exercise is that it captures only the direct effect of income on religious demand, leaving out possible indirect effects through temple supply. If lower income leads to more temples, and those temples increase affiliation, we might be understating the role of income. We return to this point in the next section and argue—based on the results presented in this section—that this is unlikely to overturn our conclusions.

for different values of $\varrho_w \in [0, 1]$, while the curves in Figure 7(b) show the average of $S_{Edt}(w_{idt}^o, N_{Cdt}^o, (1 + \varrho_N)N_{Edt}^o)$ for different values of $\varrho_N \in [0, 1]$. Because income data are unavailable for 1980, and thus pre-migration income cannot be computed for 1991, all averages are calculated using only the 2000 and 2010 Census waves.

Each panel includes two curves—one based on the model estimated with demographic controls and one without—as well as a horizontal line indicating the average Evangelical share under observed conditions, i.e., the average of $S_{Edt}(w_{idt}, N_{Cdt}, N_{Edt})$.²⁹ The intersection between each curve and this horizontal line identifies the magnitude of change in income or temple supply required, on its own, to replicate the observed effect of migration on Evangelical affiliation.

Three patterns emerge. First, the difference between each curve and the horizontal line at $\varrho_w = 0$ and $\varrho_N = 0$ reflects the variation in the Evangelical share that would have occurred in the absence of migration as predicted by the demand model, assuming migration affected outcomes only through income and Evangelical temple supply. This gap is approximately 1.29 percentage points—close to the estimated impact of migration on the Evangelical share calculated from the estimates in Table 2 for 2000 and 2010, which is 1.53 percentage points; see also Figure 4.³⁰ This suggests that although migration may have influenced religious affiliation through additional channels, income and religious infrastructure were key mechanisms.

Second, the Evangelical share is substantially more responsive to temple supply than to income: the curves in the right panel are clearly steeper, indicating stronger sensitivity to changes in infrastructure.³¹

Third, the size of the income drop or the temple expansion needed to reproduce the observed Evangelical share varies substantially between the two channels. Matching the observed Evangelical share through income alone would require a drop of 60–80%—an implausibly large decline compared to the 5% impact estimated in Table 3. By contrast, a 20% increase in Evangelical temples is sufficient to replicate the shift, closely matching the

²⁹For simplicity the horizontal line was computed using only predictions from the model estimated with demographic controls.

³⁰Recall that the counterfactual simulations are based only on the 2000 and 2010 Census waves, so we compare them to the migration effects for those years reported in Section 3.

³¹Figure F3 in the Appendix provides additional sensitivity analysis supporting this conclusion.

estimated effect of migration on infrastructure we estimated in the previous section.

Taken together, these exercises suggest that religious demand is substantially more responsive to infrastructure than to income. While Evangelical churches do appear to attract poorer individuals—consistent with prior literature (Chesnut, 1997; Mariano, 2011; Costa et al., 2019)—the magnitude of that association is modest. In contrast, the responsiveness of the demand to Evangelical temple availability is large and robust.

5.4 Robustness and Supporting Evidence

The results above rely on identifying assumptions that merit further scrutiny. The objective of this section is to assess the credibility of these assumptions and provide additional support for our main findings.

A first-order concern is that the stronger sensitivity of demand to changes in infrastructure may be driven by the endogeneity of infrastructure in the second step of our estimation procedure. First, as previously noted, the comparison between OLS and TWFE estimates in Table C8 suggests that endogeneity is unlikely to be driving the positive association between Evangelical temple density and demand. Including fixed effects—a standard approach to address unobserved heterogeneity—makes the Evangelical coefficient larger and more significant. If our estimates were upward-biased due to omitted variables correlated with both temple supply and religious demand, we would expect the opposite pattern. We interpret this as strong evidence that the estimated sensitivity of demand to infrastructure is not primarily driven by concerns of this nature.

A complementary piece of evidence on the association between Evangelical infrastructure and affiliation comes from a staggered difference-in-differences design that exploits the timing of temple entry across municipalities to further assess the validity of the estimates in Table C8. This exercise estimates the effect of temple entry directly on religious shares, rather than on the estimated demand shifters.³² We find that the entry of Evangelical temples generates a positive and increasing effect on Evangelical affiliation, with no evidence of

³²Due to the one-to-one mapping between demand shifters and market shares, the results are qualitatively comparable to those in Table C8. The sample includes only municipalities with no temples of a given denomination in 1980. For estimation, we follow Callaway and Sant’Anna (2021). See Equation (D1) in the Appendix.

pre-trends (see Figure D2 in the Appendix). In contrast, the estimated effects of Catholic temple entry are positive but very imprecise and statistically insignificant. Using a different sample and identification strategy, these results corroborate our baseline estimates: Evangelical infrastructure expansion is systematically associated with rising Evangelical affiliation, while Catholic temple entry appears unrelated to changes in Catholic shares. This reinforces the interpretation that the relationship documented in Table C8 is not driven by unobserved confounders.

Finally, we examine the dynamic response of temple supply to migration. Figure D1 in the Appendix shows that migration increases Evangelical temple density over time, but with a lag—effects only become substantial after five years. In contrast, the impact on Catholic temples is negligible at all horizons. This pattern indicates that infrastructure responds relatively slowly to demographic shocks, and that any contemporaneous correlation between temple counts and short-run demand shocks is likely to be second-order.³³

A separate concern is that the weaker relationship between income and religious demand may also be driven by income endogeneity in the first step. If income is correlated with unobserved individual characteristics—such as family background or cultural traits—that also influence religious affiliation, the estimated coefficients $\tilde{\gamma}_j$ may be biased. While we cannot fully rule out this possibility, we assess its plausibility by examining how sensitive the estimated income gradient is to the inclusion of demographic controls. Figure 7(a) shows that controlling for observables—such as education, age, and gender—reduces the estimated sensitivity of demand to income: the dashed curve (estimated without controls) lies consistently above the solid curve (with controls), indicating a weaker association once basic demographics are taken into account. If income endogeneity were the main driver of the observed relationship, we would expect the inclusion of controls to strengthen—not weaken—the negative gradient. This pattern suggests that endogeneity is unlikely to explain the limited role of income in accounting for the rise in Evangelical affiliation.

Second, one may also argue that the supply of Evangelical temples is itself a function of income: changes in income may shift religious demand, which in turn could trigger a

³³A potential problem here is that delayed temple expansion may simply reflect persistence in migration flows. However, we show that in general our results remain stable when controlling for lagged migration, which tends to attenuate this concern.

supply response. In this view, a decline in w would increase the attractiveness of Evangelical churches, leading to an expansion in N_E . As a result, the total effect of lower income on Evangelical affiliation could be stronger than what is shown in Figure 7(a), which isolates only the direct effect of w . While we admit that this concern is valid, our results strongly suggest that the demand for Evangelical churches is not very sensitive to income. This limited responsiveness implies that any indirect effect of w on N_E —and thus on religious affiliation—is likely to be small.³⁴

This observation reinforces the idea that population growth—not falling income—is the more plausible mechanism behind the expansion of Evangelical infrastructure. As shown above, demand is only weakly responsive to income, making it unlikely that the observed supply response was triggered by a shift in demand. Instead, the migration-induced increase in population expanded the local religious market, creating incentives for churches to open new temples, serving both migrants and existing residents.

In summary, while the exercises in this section rely on strong identifying assumptions, the evidence above consistently supports our main results. The core finding—that religious demand is more responsive to infrastructure than to income, and that this asymmetry underlies the counterfactual simulations—appears robust. Additional sensitivity checks in Figures 4(a) and 4(b) in the Appendix reinforce this conclusion. Importantly, potential violations of our identifying assumptions are likely to bias the estimates in directions that amplify, rather than attenuate, the relative importance of infrastructure. As a whole, these findings suggest that the expansion of Evangelical infrastructure—rather than income shocks—is the most plausible channel through which migration affected religious affiliation. Ultimately, this implies that the rise of Evangelicals may reflect a deeper institutional asymmetry: unlike the Catholic Church, Evangelical denominations are more flexible and better able to respond to

³⁴This concern stems from the fact that we do not explicitly model the supply side of religion. Doing so would require strong assumptions and pose significant challenges. First, defining geographic markets is particularly difficult in large metropolitan areas like São Paulo and Rio de Janeiro. Second, the number of religious organizations is vast and highly heterogeneous, complicating the structure and solution of any equilibrium model. Third, the objective function of churches may not be well-defined or easily modeled. These limitations prevent a precise understanding of the origins of the observed supply-side response, which we interpret—given the limited role of income in shaping demand—as a consequence of population growth induced by migration. While this interpretation is consistent with the reduced-form evidence, it is not derived directly from the model.

demographic shocks.

6 Conclusions

This paper provides new evidence that internal migration was a key driver of Brazil’s Evangelical transformation. Exploiting exogenous variation in rural-to-urban migration flows induced by commodity price shocks, we show that migration increased Evangelical affiliation and reduced Catholic affiliation across Brazilian municipalities.

We identify two mechanisms through which this effect operates. First, migration depressed local incomes, which may have raised the relative appeal of Evangelical churches given their outreach strategies in low-income settings. Second, and more importantly, migration led to an asymmetric expansion in religious infrastructure. Evangelical denominations responded aggressively to population growth by increasing the number of temples, while the Catholic Church did not. This pattern reflects institutional differences in organizational structure, cost, and responsiveness to local market conditions.

To quantify the relative importance of these channels, we developed and estimated a structural model of religious choice. Counterfactual simulations based on this model show that supply-side expansion alone accounts for most of the observed religious shift, suggesting that in contexts of rapid urbanization and population growth, the capacity to expand infrastructure—and to do so quickly—can prove decisive in shaping long-run equilibria in religious markets.

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Appendix

This appendix provides supporting material and robustness checks for the main findings in the paper. Section A documents the institutional background and the classification of religious denominations in Brazil used throughout the analysis. Section B presents robustness checks for the reduced-form SSIV results. Section C reports the structural estimates for the determinants of religious demand. Section D includes additional figures and tables illustrating dynamic and difference-in-differences supply-side effects. Section E provides technical details on how we construct the counterfactual simulations reported in the main text. Section F provides additional sensitivity analysis of religious demand with respect to changes in income and temple supply.

A Institutional Background Details

In this appendix, we provide a detailed explanation of the classification of religious denominations in Brazil, particularly the composition of the Evangelical group as used in this study. Table A1 presents the main religious traditions and the denominations within each category. This classification builds upon prior work, notably Mariano (2014), and offers a structured taxonomy for analyzing religious affiliation in Brazil.

The largest religious tradition in Brazil is the Catholic Church (Group I), which primarily consists of the Roman Catholic Church. Although the Orthodox Church is present in Brazil, its adherents are relatively few and do not significantly impact the broader religious landscape.

The Evangelical denominations are categorized into three distinct groups: Mainline Protestant (II), Non-Pentecostal Evangelical (III), and Pentecostal Evangelical (IV), with a separate category for Non-denominational Evangelicals (V). The classification of Evangelical denominations is not always clear-cut, as theological and institutional differences can blur traditional boundaries. Nonetheless, this taxonomy provides a useful framework for distinguishing major branches.

Mainline Protestant churches (II) include Lutheran, Anglican, Calvinist, and Anabaptist traditions. These groups are historically connected to the Protestant Reformation and were introduced to Brazil primarily through European immigration. These churches tend to have more hierarchical structures and formalized doctrines compared to other Evangelical groups.

Non-Pentecostal Evangelicals (III) include denominations such as Presbyterians, Congregationalists, Baptists, Methodists, and Adventists. Many of these groups were influenced by the religious revivals of the 18th and 19th centuries, particularly the Second Great Awakening in the United States. They emphasize personal conversion experiences, scriptural authority, and active missionary work.

Pentecostal Evangelicals (IV) encompass a broad array of denominations, many of which emerged in Brazil during the 20th century. These churches emphasize direct experiences with the Holy Spirit, such as speaking in tongues, divine healing, and prophecy. They are often further divided into Classical Pentecostals, Neo-Pentecostals, and newer Charismatic movements. Notable denominations in this category include the Assemblies of God, Universal Church of the Kingdom of God, and the Foursquare Church. Neo-Pentecostalism,

in particular, is characterized by its embrace of the prosperity gospel and highly dynamic worship services (Zilla, 2018).

Table A1: **Religious Denominations in Brazil**

	Group	Denominations
I	Catholic	Roman Catholic
II	Mainline Protestant	Lutheran, Anglican, Calvinist, Anabaptist
III	Non-Pentecostal Evangelical	Presbyterian, Congregationalist, Baptist, Methodist, Adventist
IV	Pentecostal Evangelical	Christian Congregation, Assembly of God, Foursquare, Universal Church of the Kingdom of God, Brasil para Cristo, Deus é Amor, Renascer, Mundial, Nazareno, Casa da Benção, Casa da Oração, Maranata, Igreja da Graça
V	Non-denominational Evangelicals	Various minor Evangelical denominations following hybrid (Pentecostal and Non-Pentecostal) lines
VI	Other	Afro-Brazilian, Spiritism, Eastern Religions, Judaism, Islam

Non-denominational Evangelicals (V) include various independent churches that do not align strictly with Pentecostal or traditional Protestant categories. These groups often blend elements from different theological traditions and exhibit a high degree of organizational flexibility. Their growth has been fueled by urbanization and the increasing individualization of religious practice.

Lastly, the category “Other” includes religious traditions outside of Christianity, such as Afro-Brazilian religions (e.g., Candomblé and Umbanda), Spiritism, Eastern religions, Judaism, and Islam. These groups, while historically significant, represent a relatively small fraction of the Brazilian population.

B Robustness Checks

This Appendix presents a comprehensive set of robustness checks for our reduced-form SSIV estimates. We examine the effects of migration on Catholic and Evangelical shares (Tables B2 and B3), on labor market outcomes (Table B4), and on the number of temples for each religious group (Tables B5 and B6). These results verify that our findings are robust to a range of controls and specifications.

Table B2: **Effects of Migration on the Share of Catholics among Native Populations**

	Dep. Var.: Δ share of catholics						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<u>Panel A: Δ 1980-1991</u>							
In-migration rate	-0.118* (0.069)	-0.109* (0.063)	-0.108* (0.063)	-0.109* (0.063)	-0.121*** (0.030)		
First Stage F Stat	185.9	186.4	176.6	186.4	343.9		
Observations	3,989	3,989	3,989	3,989	3,989		
<u>Panel B: Δ 1991-2000</u>							
In-migration rate	-0.181*** (0.026)	-0.167*** (0.020)	-0.166*** (0.019)	-0.167*** (0.020)	-0.122*** (0.018)	-0.166*** (0.020)	-0.081*** (0.022)
First Stage F Stat	192.2	199.1	189.7	210.5	267.6	183.0	103.6
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
<u>Panel C: Δ 2000-2010</u>							
In-migration rate	-0.120*** (0.018)	-0.103*** (0.014)	-0.103*** (0.014)	-0.103*** (0.014)	-0.078*** (0.015)	-0.093*** (0.014)	-0.042*** (0.010)
First Stage F Stat	70.9	69.0	68.6	69.5	160.4	64.6	58.7
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
Baseline covariates		✓	✓	✓	✓	✓	✓
Lagged share of catholics		✓	✓	✓	✓	✓	✓
Dest. price shocks			✓				
Price dist. weighted				✓			
State dummies					✓		
Lagged dep. var.						✓	
Lagged migration							✓

This table shows origin-level SSIV coefficients on the 5-year accumulated changes in the municipality-level share of catholics. The SSIV is the inflow of migrants predicted by price shocks at the origin municipality. In Panel A, the dependent variable is the change in the outcome between 1980 and 1991, while in Panels B and C is the change between 1991-2000 and 2000-2010, respectively. Columns 2-7 control for lagged covariates (share of population less than 18 years old; share of non-white; share of female; share with complete high-school or higher; share of married people in the population; household income per capita, and the total population) as well as the lagged share of Catholics. Column 3 also controls for the price shocks at the destination, column 4 includes price shocks for each destination municipality, weighted by the inverse distance between each origin-destination pair, column 5 includes a dummy for each state, column 6 also controls for the lagged dependent variable, and column 7 controls for the lagged log of the predicted migration. All regressions are weighted by the total population in 1980. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table B3: **Effects of Migration on the Share of Evangelicals among Native Populations**

	Dep. Var.: Δ share of evangelicals						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<u>Panel A: Δ 1980-1991</u>							
In-migration rate	0.041* (0.023)	0.046* (0.025)	0.047* (0.026)	0.047* (0.025)	0.043*** (0.011)		
First Stage F Stat	185.9	181.6	171.9	181.2	347.5		
Observations	3,989	3,989	3,989	3,989	3,989		
<u>Panel B: Δ 1991-2000</u>							
In-migration rate	0.152*** (0.031)	0.148*** (0.030)	0.146*** (0.030)	0.144*** (0.030)	0.098*** (0.015)	0.143*** (0.029)	0.060* (0.034)
First Stage F Stat	192.2	186.6	176.2	193.6	280.7	182.5	106.1
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
<u>Panel C: Δ 2000-2010</u>							
In-migration rate	0.083*** (0.012)	0.073*** (0.010)	0.072*** (0.009)	0.073*** (0.009)	0.052*** (0.011)	0.041*** (0.008)	0.025*** (0.007)
First Stage F Stat	70.9	68.4	68.1	68.7	173.6	54.3	61.2
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
Baseline covariates		✓	✓	✓	✓	✓	✓
Lagged share of evangelicals		✓	✓	✓	✓	✓	✓
Dest. price shocks			✓				
Price dist. weighted				✓			
State dummies					✓		
Lagged dep. var.						✓	
Lagged migration							✓

This table shows origin-level SSIV coefficients on the 5-year accumulated changes in the municipality-level share of evangelicals. The SSIV is the inflow of migrants predicted by price shocks at the origin municipality. In Panel A, the dependent variable is the change in the outcome between 1980 and 1991, while in Panels B and C is the change between 1991-2000 and 2000-2010, respectively. Columns 2-7 control for lagged covariates (share of population less than 18 years old; share of non-white; share of female; share with complete high-school or higher; share of married people in the population; household income per capita, and the total population) as well as the lagged share of Evangelicals. Column 3 also controls for the price shocks at the destination, column 4 includes price shocks for each destination municipality, weighted by the inverse distance between each origin-destination pair, column 5 includes a dummy for each state, column 6 also controls for the lagged dependent variable, and column 7 controls for the lagged log of the predicted migration. All regressions are weighted by the total population in 1980. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table B4: **Effects of Migration on Earnings of Native Workers**

	Dep. Var.: Δ log-earnings						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<u>Panel A: Δ 1991-2000</u>							
In-migration rate	-0.344*** (0.095)	-0.401*** (0.092)	-0.404*** (0.093)	-0.396*** (0.092)	-0.303*** (0.051)	-0.300*** (0.070)	-0.296** (0.135)
First Stage F Stat	406.5	192.2	184.0	198.7	297.1	173.1	107.5
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
<u>Panel B: Δ 2000-2010</u>							
In-migration rate	-0.402*** (0.126)	-0.540*** (0.199)	-0.540*** (0.197)	-0.540*** (0.199)	-0.256*** (0.092)	-0.628*** (0.237)	-0.270 (0.197)
First Stage F Stat	227.7	70.9	70.9	71.1	191.9	73.3	63.0
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
Baseline covariates		✓	✓	✓	✓	✓	
Lagged share of evangelicals		✓	✓	✓	✓	✓	
Dest. price shocks			✓				
Price dist. weighted				✓			
State dummies					✓		
Lagged dep. var.						✓	
Lagged migration							✓

This table shows origin-level SSIV coefficients on the 5-year accumulated changes in the municipality-level log-earnings for native workers. The SSIV is the inflow of migrants predicted by price shocks at the origin municipality. In Panel A, the dependent variable is the change in the outcome between 1991-2000, while in Panel B is the change between 2000-2010. Columns 2-6 control for lagged covariates (share of population less than 18 years old; share of non-white; share of female; share with complete high-school or higher; share of married people in the population; household income per capita, and the total population). Column 3 also controls for the price shocks at the destination, column 4 includes price shocks for each destination municipality, weighted by the inverse distance between each origin-destination pair, column 5 includes a dummy for each state, column 6 also controls for the lagged dependent variable, and column 7 controls for the lagged log of the predicted migration. All regressions are weighted by the total population in 1980. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table B5: **Effects of Migration on the Number of Catholic Temples**

	Dep. Var.: Δ number of catholic temples						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<u>Panel A: Δ 1980-1991</u>							
In-migration rate	0.068*** (0.013)	0.079*** (0.016)	0.076*** (0.016)	0.079*** (0.016)	0.070*** (0.013)		
First Stage F Stat	326.8	185.9	177.7	185.2	366.9		
Observations	3,989	3,989	3,989	3,989	3,989		
<u>Panel B: Δ 1991-2000</u>							
In-migration rate	-0.002 (0.010)	-0.021 (0.015)	-0.023 (0.016)	-0.021 (0.015)	-0.057*** (0.016)	-0.018 (0.016)	0.012 (0.020)
First Stage F Stat	406.5	192.2	184.0	198.7	297.1	194.5	107.5
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
<u>Panel C: Δ 2000-2010</u>							
In-migration rate	-0.003 (0.009)	-0.006 (0.014)	-0.007 (0.014)	-0.006 (0.014)	0.010 (0.010)	-0.006 (0.014)	0.033** (0.013)
First Stage F Stat	227.7	70.9	70.9	71.1	191.9	76.6	63.0
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
Baseline covariates		✓	✓	✓	✓	✓	
Dest. price shocks			✓				
Price dist. weighted				✓			
State dummies					✓		
Lagged dep. var.						✓	
Lagged migration							✓

This table shows origin-level SSIV coefficients on the 5-year accumulated changes in the municipality-level number of catholic temples. The SSIV is the inflow of migrants predicted by price shocks at the origin municipality. In Panel A, the dependent variable is the change in the outcome between 1980 and 1991, while in Panels B and C is the change between 1991-2000 and 2000-2010, respectively. Columns 2-6 control for lagged covariates (share of population less than 18 years old; share of non-white; share of female; share with complete high-school or higher; share of married people in the population; household income per capita, and the total population). Column 3 also controls for the price shocks at the destination, column 4 includes price shocks for each destination municipality, weighted by the inverse distance between each origin-destination pair, column 5 includes a dummy for each state, column 6 also controls for the lagged dependent variable, and column 7 controls for the lagged log of the predicted migration. All regressions are weighted by the total population in 1980. Standard errors clustered at the origin municipality level in parentheses.

*** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table B6: **Effects of Migration on the Number of Evangelical Temples**

	Dep. Var.: Δ number of evangelical temples						
	(1)	(2)	(3)	(4)	(5)	(6)	(7)
<u>Panel A: Δ 1980-1991</u>							
In-migration rate	0.863*** (0.117)	0.883*** (0.151)	0.882*** (0.151)	0.883*** (0.151)	0.858*** (0.091)		
First Stage F Stat	326.8	185.9	177.7	185.2	366.9		
Observations	3,989	3,989	3,989	3,989	3,989		
<u>Panel B: Δ 1991-2000</u>							
In-migration rate	0.748*** (0.115)	0.685*** (0.128)	0.686*** (0.129)	0.688*** (0.128)	0.565*** (0.077)	0.538*** (0.156)	0.419*** (0.160)
First Stage F Stat	406.5	192.2	184.0	198.7	297.1	88.2	107.5
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
<u>Panel C: Δ 2000-2010</u>							
In-migration rate	0.737*** (0.178)	0.651** (0.255)	0.654*** (0.247)	0.651** (0.255)	0.516*** (0.101)	0.360 (0.230)	0.284 (0.227)
First Stage F Stat	227.7	70.9	70.9	71.1	191.9	34.9	63.0
Observations	3,989	3,989	3,989	3,989	3,989	3,989	3,989
Baseline covariates		✓	✓	✓	✓	✓	
Dest. price shocks			✓				
Price dist. weighted				✓			
State dummies					✓		
Lagged dep. var.						✓	
Lagged migration							✓

This table shows origin-level SSIV coefficients on the 5-year accumulated changes in the municipality-level number of evangelical temples. The SSIV is the inflow of migrants predicted by price shocks at the origin municipality. In Panel A, the dependent variable is the change in the outcome between 1980 and 1991, while in Panels B and C, it is the change between 1991-2000 and 2000-2010, respectively. Columns 2-6 control for lagged covariates (share of population less than 18 years old; share of non-white; share of female; share with complete high-school or higher; share of married people in the population; household income per capita, and the total population). Column 3 also controls for the price shocks at the destination, column 4 includes price shocks for each destination municipality, weighted by the inverse distance between each origin-destination pair, column 5 includes a dummy for each state, column 6 also controls for the lagged dependent variable, and column 7 controls for the lagged log of the predicted migration. All regressions are weighted by the total population in 1980. Standard errors clustered at the origin municipality level in parentheses. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

C Structural Model Estimates

This Appendix reports the main structural estimates used in our demand model. Table C7 presents the estimated effects of income on the demand for Catholic and Evangelical affiliation. Table C8 reports the relationship between the number of temples and the estimated demand shifters $\tilde{\delta}_{jd}$ across years and specifications, using both OLS and TWFE estimators.

Table C7: **Effects of Earning Variations on the Demand for Churches**

	(1)	(2)	(3)	(4)	(5)	(6)
	1991		2000		2010	
	Catholic	Evangelical	Catholic	Evangelical	Catholic	Evangelical
Panel A: With Demographic Controls						
Income	-0.00485 (0.00845)	-0.166*** (0.0229)	0.0708*** (0.0158)	-0.109*** (0.0215)	0.0110 (0.0103)	-0.144*** (0.0134)
Observations	1,735,871	1,735,871	2,481,561	2,481,561	3,123,037	3,123,037
Panel B: Without Demographic Controls						
Income	-0.0214*** (0.00649)	-0.161*** (0.0249)	0.00696 (0.00559)	-0.120*** (0.0163)	-0.00142 (0.00201)	-0.106*** (0.0109)
Observations	1,735,871	1,735,871	2,301,341	2,301,341	2,956,642	2,956,642
Municipality Dummies	✓	✓	✓	✓	✓	✓

This table presents estimates of the effect of income on religious demand, following equation (12). The model is estimated using Maximum Likelihood for samples of individuals from the Brazilian Census for the corresponding years. Individuals' choice sets include three options: Catholic, Evangelical, and no religion. Panel A includes demographic controls (education, age profile, and gender dummies), while Panel B does not. All regressions include municipality fixed effects. Standard errors are clustered at the municipality level. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Table C8: **Effects of Number of Temples on the Demand Shifters $\tilde{\delta}_{jd}$**

	(1)	(2)	(3)	(4)
	No Demographic Controls		Demographic Controls	
	Evangelical	Catholic	Evangelical	Catholic
<u>Panel A: Pooled OLS</u>				
Number of Temples $\times$ Year = 1991	0.264 (0.419)	7.097** (2.828)	0.170 (0.405)	6.783** (2.752)
Number of Temples $\times$ Year = 2000	0.000749 (0.234)	6.114*** (1.083)	0.0855 (0.224)	5.720*** (0.976)
Number of Temples $\times$ Year = 2010	-0.208 (0.197)	8.294*** (0.840)	-0.150 (0.186)	8.059*** (0.811)
<u>Panel B: TWFE</u>				
Number of Temples $\times$ Year = 1991	0.916*** (0.322)	1.926 (1.957)	0.792** (0.316)	1.858 (1.957)
Number of Temples $\times$ Year = 2000	0.926*** (0.310)	-1.368* (0.786)	0.916*** (0.306)	-1.510* (0.820)
Number of Temples $\times$ Year = 2010	0.726*** (0.232)	-0.754 (0.781)	0.688*** (0.229)	-0.717 (0.777)
Observations	735	735	671	671

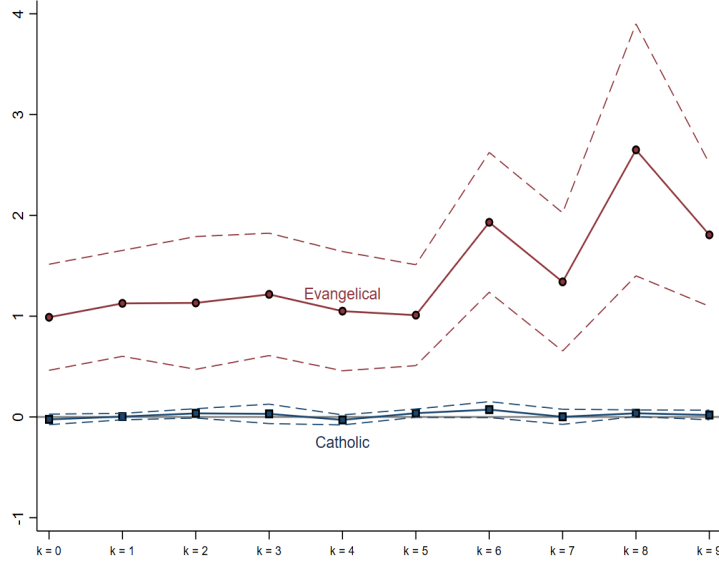
This table reports pooled OLS and two-way fixed effects (TWFE) estimates of the effect of the number of temples on the demand shifters, $\tilde{\delta}_{jd}$, for each religious group. All regressions include year fixed effects, and TWFE specifications additionally include municipality fixed effects. Regressions are estimated separately for each group. The coefficients on the number of temples are interacted with year dummies. In models with demographic controls, the dependent variable $\tilde{\delta}_{jd}$ is obtained from a first-stage regression that includes income, education, age profile, and gender dummies. Models without demographic controls exclude education, age profile, and gender dummies. Standard errors, clustered at the municipality level, are reported in parentheses.

D Additional Tables and Figures

This Appendix presents additional graphical evidence on the dynamic effects of migration and temple entry. Figure D1 illustrates how migration shocks affect the entry of temples over time, while Figure D2 reports staggered difference-in-differences estimates of the effect of temple entry on religious affiliation.

Dynamic effects of migration. The exercise in Figure D1 is based on SSIV regressions of $N_{jd,t+k} - N_{jd,t+k-1}$ on m_{dt} , with $k \in 0, 1, 2, \dots, 9$. For each value of k , a separate regression is estimated for each religious group, pooling data across all Census years. Intuitively, the coefficient on m_{dt} for a given k captures the average effect—across municipalities and Census waves—of a migration shock at time t on the change in the stock of temples of each religious group k periods ahead. All regressions include the same baseline controls used in Table 4, along with year fixed effects. The coefficients plotted in Figure D1 represent the estimated impact of m_{dt} on $N_{jd,t+k} - N_{jd,t+k-1}$ for each horizon k . The results suggest that migration has a stronger effect on investments in Evangelical temples in the longer run. *** Significant at 1%. ** Significant at 5%. * Significant at 10%.

Figure D1: **Effects of migration on the dynamic entry of temples**



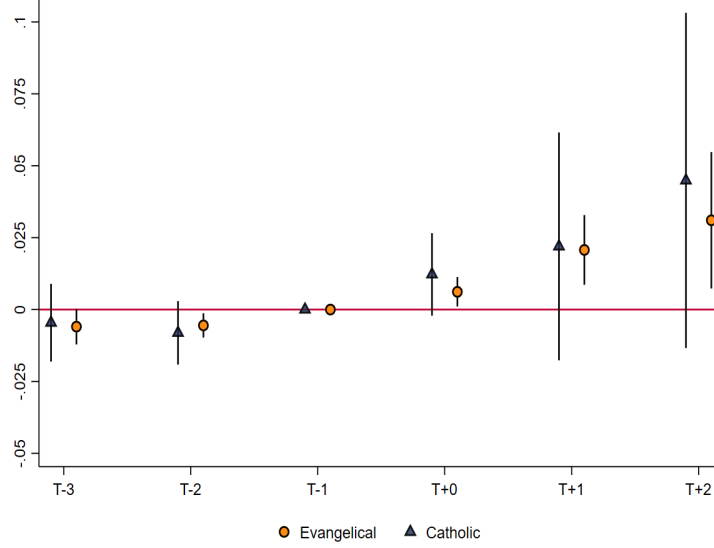
This figure shows the dynamic effects of in-migration on the entry of Catholic and Evangelical temples. Each point is an SSIV coefficient for a regression of a contemporaneous shock of incoming migration on changes in the number of temples per capita k years ahead. Dashed lines represent the 95% confidence interval.

Differences-in-Differences for temple entry. The results in Figure D2 are based on the estimation of the following equation:

$$Y_{it} = \alpha_i + \beta_t + \sum_{\substack{h=-a \\ h \neq -1}} \tau_h \mathbb{1}[K_{it} = h] + \tau_b \mathbb{1}[K_{it} \geq b] + \varepsilon_{it}, \quad (\text{D1})$$

where, α_i and β_t are municipality and time fixed effects; $a \geq 0$ and $b \geq 0$ are the included leads and lags of the treatment, and ε_{it} is the error term. We define a municipality as treated if, between Census years t and $t + 1$, at least one temple began operating.

Figure D2: **Effects of the Dynamic Entry of Temples on Religious Affiliation**



This figure shows the dynamic effects of temple entry on the proportion of the population affiliated with each religious group, using the estimator developed in Callaway and Sant’Anna (2021). For each religious group, the sample includes only municipalities with no temples of that denomination operating in 1980. There are 2,445 municipalities without Evangelical temples in 1980 and 2,100 without Catholic temples, corresponding to 61.7% and 53.0% of the original sample, respectively. Vertical lines on each coefficient represent 95% confidence intervals.

E Counterfactual Details

This Appendix provides full details on the construction of counterfactual variables used in our simulations. It explains how we construct the counterfactual average income in the absence of migration (w_{id}^o) and the counterfactual number of temples (N_{jd}^o) using the estimated coefficients from the reduced-form analysis. These quantities underpin the simulations discussed in Section 5 of the main text.

To construct the counterfactuals in Section 5, we estimate two key components: (i) the counterfactual average income in the absence of migration, denoted w_{id}^o , and (ii) the counterfactual number of Evangelical and Catholic temples, denoted N_{jd}^o , where $j = E$ for Evangelicals and $j = C$ for Catholics.

To compute w_{id}^o , we proceed in three steps. First, using the estimates from column (1) of Table 3, we calculate the counterfactual income growth rate for each municipality and period as

$$g_{dt}^0 = \Delta \ln(w_{dt}) - \hat{\beta}_t m_{dt},$$

where $\hat{\beta}_t$ is the estimated effect of migration on income growth for period t , $\ln(w_{dt})$ is the log of average income in municipality d and period t , and $\Delta \ln(w_{dt})$ is the observed change

in log income; i.e. the dependent variable in the equation that we used to estimate β . The term g_{dt}^0 captures how income would have evolved in the absence of migration.

Second, we use this growth rate to construct the counterfactual average income in the municipality as

$$w_{dt}^0 = w_{dt-1}(1 + g_{dt}^0),$$

where w_{dt-1} is the observed average income in the previous period.

Third, to obtain individual-level counterfactual income, we regress observed individual income on municipality dummies (separately for each period) and collect the residuals, denoted η_{idt} . We then define:³⁵

$$w_{id}^o = w_{dt}^0 + \eta_{idt}.$$

The construction of the counterfactual number of temples N_{jd}^o follows a similar logic. First, using the estimates from columns (2) and (4) of Table 4, we compute

$$h_{jdt}^0 = \Delta N_{jdt} - \hat{\gamma}_{jt} m_{dt},$$

where $\hat{\gamma}_{jt}$ is the estimated effect of migration on the number of temples of religious group j in period t , and ΔN_{jdt} is the observed variation in the number of temples. This gives us the change in temple count that would have occurred without migration. Then we compute the counterfactual number of temples as

$$N_{jdt}^o = N_{jdt-1} + h_{jdt}^0,$$

where N_{jdt-1} is the observed stock of temples in the previous period.

F More on the Sensitivity of the Demand to Income and Infrastructure

This Appendix presents additional evidence on the sensitivity of religious demand to changes in income and infrastructure. While Figure 6 offers useful insight, it is based on pre-migration counterfactuals, excludes the 1991 Census, and considers only one direction of variation—increasing income and reducing temple supply. Here, we adopt a more straightforward approach designed to infer the sensitivity of religious demand under different conditions.

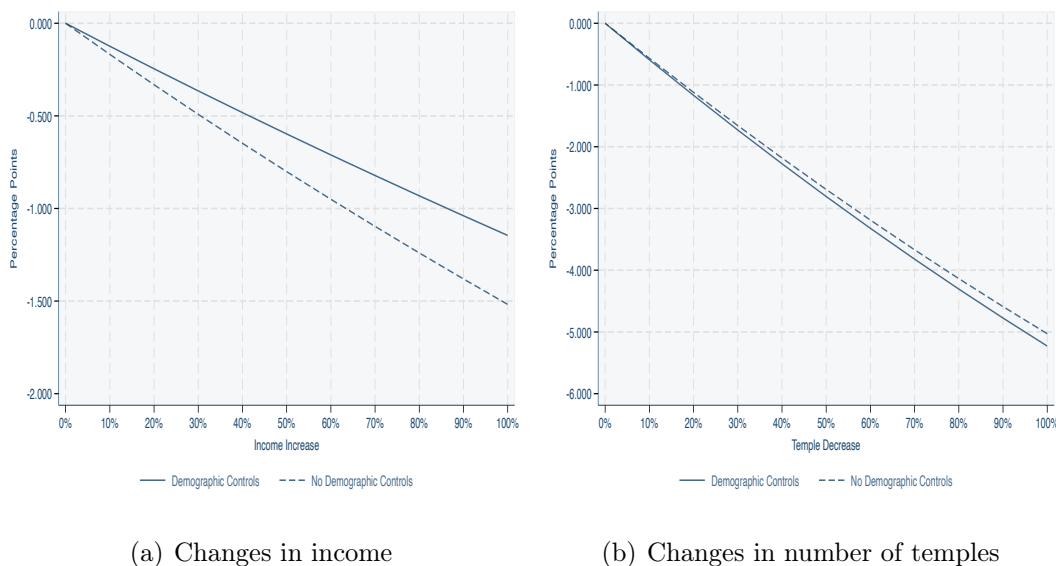
We begin by varying income. For each year (1991, 2000, 2010), we simulate proportional increases in individuals' observed income—unlike Figure 6, which considers only 2000 and 2010, uses pre-migration counterfactuals, and focuses on income declines. Specifically, we compute predicted changes in the Evangelical share as income increases by a factor $\varrho \in [1, 2]$, with $\varrho = 1$ denoting baseline income and $\varrho = 2$ a doubling. For each individual, we calculate the change in predicted Evangelical affiliation across income levels and average these differences using Census sampling weights. Figure 4(a) reports the results for models

³⁵To ensure comparability, we define observed income as $w_{id} = w_{dt} + \eta_{idt}$, using the same residual η_{idt} as in the counterfactual $w_{id}^o = w_{dt}^0 + \eta_{idt}$. This approach holds individual heterogeneity fixed and attributes the effect of migration solely to changes in the municipal income component.

estimated with and without demographic controls. In all cases, higher income is associated with lower Evangelical affiliation. For instance, doubling income reduces the Evangelical share by approximately 1.1 percentage points when demographic controls are included.

Next, we conduct a parallel simulation for religious infrastructure. Using the same framework, we simulate the effect of reducing the number of observed Evangelical temples by a factor $\varrho \in [0, 1]$, where $\varrho = 0$ corresponds to baseline temple supply and $\varrho = 1$ to complete removal. As in the income counterfactuals, we compute the predicted change in Evangelical affiliation for each individual and average across the sample using Census weights. Figure 4(b) reports the results. The decline in Evangelical affiliation is large and monotonic: eliminating half of all temples leads to a drop of about 2 percentage points in the Evangelical share, and full removal results in declines exceeding 4 percentage points. Notably, these effects are substantially larger than those observed in the income simulations. The observed patterns are similar across model specifications.

Figure F3: **Sensitivity of Evangelical Affiliation to Changes in Income and Infrastructure**



Panel (a) illustrates the sensitivity of the share of Evangelicals to proportional increases in income, with $\varrho \in [1, 2]$, where $\varrho = 1$ denotes baseline income and $\varrho = 2$ represents a doubling. Panel (b) shows the sensitivity to proportional decreases in the number of temples, with $\varrho \in [0, 1]$, where $\varrho = 0$ is the baseline and $\varrho = 1$ indicates complete removal of temples. In both panels, the y-axis reports the average change in Evangelical affiliation (in percentage points) *relative to the baseline scenario*—i.e., the predicted difference between each counterfactual and the observed data. The solid line corresponds to models estimated with demographic controls; the dashed line excludes them.